# Computer Science — Class 7

Easy Exam Notes • Unit 2 to Unit 6

**UNITS IN THESE NOTES**

| # | Unit |
|---|---|
| 2 | Digital Skills |
| 3 | Computational Thinking |
| 4 | Programming |
| 5 | Digital Citizenship |
| 6 | Entrepreneurship in Digital Age |

5 Units • Matched topic-by-topic against the official PCTB textbook

**HOW TO READ EVERY UNIT**

1 Topic notes › 2 Exercise Long Questions with full answers › 3 Short Questions › 4 MCQs with the correct answer marked › 5 Mistakes found in the book

- ✔ Every topic and sub-topic in the book is covered — checked page by page against the textbook.
- ✔ Every official Exercise Long Question, Short Question and MCQ is answered, with extra practice questions added.
- ✔ Written in short, simple sentences in easy English, so every student can read it once and remember it.

Prepared by Ali Hamza, Subject Teacher — Computer Science

# Unit 2 — Digital Skills

## TOPIC 1: WORD PROCESSOR

A **word processor** is a program on a computer that lets us type, change, save and print text.

### Purpose of a word processor (2.1)

A word processor helps us make documents like letters, notes, reports and homework. It has four main jobs:

1. **Composing:** Typing new text on the page.
2. **Editing:** Changing existing text by adding, removing or fixing words.
3. **Saving:** Storing the document so it can be opened again later.
4. **Printing:** Sending the document to a printer for a paper copy.

### Word processing software (2.1.1)

A word processor lets us type, format, save and print text (words and numbers).

**Microsoft Word (MS Word)** is the most popular word processor. People use it to make brochures, letters, class activities, quizzes, tests and homework.

**Examples of word processors:** Google Docs, WordPerfect, LibreOffice Writer, Dragon, Scrivener and RagTime — even Notepad and Open Office can be used to write documents.

### Open a new Word document (2.1.2)

A new document usually starts blank, but we can also start from a ready-made design called a **template**:

1. Open Word. If it's already open, click **File > New**.
2. In the "Search for online templates" box, type a word like *letter*, *resume* or *invoice*, or pick a category such as Business, Personal or Education.
3. Click a template to preview it.
4. Click **Create**, and the new document opens.

### Interface of MS Word (2.1.3)

The **interface** is the screen we see when MS Word opens. Its main parts are:

1. **Title Bar:** At the top; shows the document's name, like "Document1 - Word".
2. **Quick Access Toolbar:** Top-left corner; frequently used buttons like Save, Undo and Redo.
3. **Tabs:** File, Home, Insert, Design, Layout, References, Mailings, Review and View. Each opens its own set of commands.
4. **Ribbon:** The wide strip below the tabs, showing commands for the open tab, grouped under names like Clipboard, Font, Paragraph and Styles.
5. **Help Button:** Type what we want to do into the "Tell me" box to find it.
6. **Window Controls:** The Minimize, Maximize and Close buttons, in the top-right corner.
7. **Scroll Bars:** The vertical one moves the page up and down; the horizontal one moves it left and right.
8. **Zoom in / Zoom out:** In the bottom-right corner; makes the page look bigger or smaller.

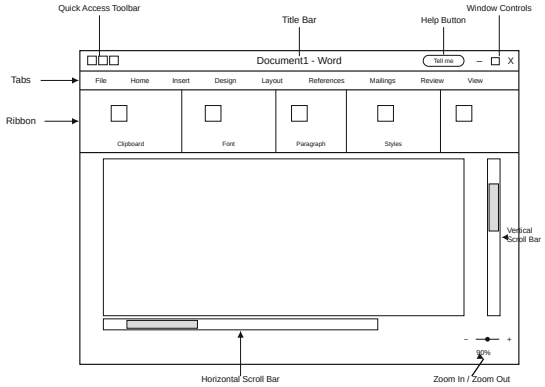



*Fig: The MS Word 2016 interface — Title Bar, Quick Access Toolbar, Tabs, Ribbon, Help Button, Window Controls, Scroll Bars and Zoom*

**Long Question: Describe the purpose of a word processor.** *[Exercise — Brief Q7]*

A word processor's purpose is to let us type, edit, format, save and print documents like letters, reports and homework.

1. **Meaning:** A word processor is a computer program that lets us enter, format, save and print text. Microsoft Word is the most popular one.
2. **Composing:** Typing new text, such as a letter, an application or homework.
3. **Editing:** Mistakes are easy to fix by adding or deleting words, without rewriting the page.
4. **Formatting:** We can make text bold, italic or underlined, and change its font, size, colour or alignment.
5. **Saving:** Lets us save the document and open it again later.
6. **Printing:** A printer gives us a paper copy.
7. **Uses:** Letters, brochures, quizzes, tests, reports and homework.

## TOPIC 2: WORKING WITH TEXT IN MS WORD

MS Word has many tools a simple text editor lacks, such as changing how text looks and adding pictures.

### Adding and modifying the text (2.1.4)

**Character formatting** is the look given to text: font, font size, bold, italic, underline, strikethrough, subscript, superscript, font colour and highlight colour.

1. **Font Face:** Changes the shape of the letters.
2. **Font Size:** Makes the text bigger or smaller.
3. **Font Colour:** Changes the colour of the text.

To format any text, we first **select** it, then use the tools on the Home tab.

There are three font styles on the Home tab:

1. **Bold (B):** Makes text look thick and dark.
2. **Italic (I):** Makes text lean (slant) to one side.
3. **Underline (U):** Draws a line under the selected text.

**Save and open files (2.1.5)**

After writing text, we save the document so we can open and use it again later.

1. **To Save:** Go to the **File** tab and click **Save**, or click the Save button on the Quick Access Toolbar, then pick a folder and name for the document.
2. **To Open:** Go to the **File** tab and click **Open**, then pick the saved file.

**Format a document (2.1.6)**

- **Text Alignment:** Where text sits on the page. MS Word has **four** types of alignment:
  1. **Left Align** (Ctrl + L)
  2. **Center Align** (Ctrl + E)
  3. **Right Align** (Ctrl + R)
  4. **Justify** (Ctrl + J): text touches both the left and right margins.

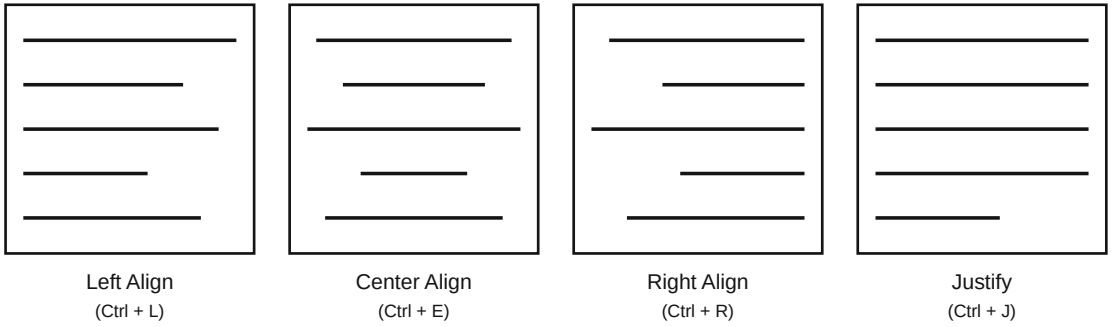



*Fig: The four types of text alignment in MS Word — Left, Center, Right and Justify*

- **Page Margins:** The blank spaces on the top, bottom, left and right of a page, which make a document look neat. Change them from the **Margins** button on the **Layout** tab.
- **Page Layout:** The **Layout** tab controls how the whole document looks. It lets us:
  1. Apply themes and colour schemes.
  2. Change the **orientation**: Portrait (tall) or Landscape (wide).
  3. Change page size, margins, indentation, line spacing and paragraph settings.

**Remember:** Portrait is a tall page; Landscape is a wide page.

## TOPIC 3: PICTURES, CUT, COPY, PASTE AND SPELL CHECK

MS Word lets us add pictures to a document and move text easily from one place to another.

**Insert, resize and add text in images (2.1.7)**

**To insert an image:**

1. Put the cursor where you want the image.
2. Click the **Insert** tab.
3. Click **Pictures** in the Illustrations group.

4. Pick where the picture comes from:
    - **This Device:** a picture saved on your computer.
    - **Stock Images:** ready-made icons, stickers and pictures.
    - **Online Pictures:** search pictures using Bing (Microsoft's search engine), which also links to OneDrive.
5. Find the picture in the Insert Picture box.
6. Click **Insert**.

**To resize an image:** Click the image to show a border with handles, then drag a handle to resize it.

**To align an image with text:** Click the image, then click **Layout Options**, and pick one: In Line with Text, Square, Tight, Through, Top and Bottom, Behind Text, or In Front of Text.

### Cut, copy and paste (2.1.8)

1. **Cutting:** Removes text or an image from its place and stores it in the **clipboard**, a temporary storage spot in the computer.
2. **Copying:** Makes a copy of text or an image in the clipboard, while the original stays in place.
3. **Pasting:** Moves the text or image from the clipboard to a new place in the document.

**Keyboard shortcuts:**

| Command | Shortcut | What it does |
|---|---|---|
| Cut | Ctrl + X | Removes the selected item and keeps it in the clipboard |
| Copy | Ctrl + C | Makes a copy of the selected item in the clipboard |
| Paste | Ctrl + V | Puts the item from the clipboard where the cursor is |

**Ribbon method:** Select the text or image, go to the **Home** tab, then click Cut or Copy in the Clipboard group. This needs more steps than shortcuts but gives more paste choices.

### Spell check (2.1.9)

Spell check is a tool built into MS Word that checks our spelling and grammar mistakes.

1. If it's turned on, Word underlines spelling and grammar mistakes automatically while we type.
2. We can also open the **Spelling & Grammar** box from the **Review** tab, which checks all mistakes at once.
3. The shortcut key for spelling and grammar check is **F7**.

## TOPIC 4: LISTS, SAVE VS SAVE AS, TABLES, HEADER/FOOTER, THESAURUS AND PRINTING

These tools help us arrange our document, making it look neat and professional.

### Create and modify numbered and bulleted lists (2.1.10)

1. **Numbered List:** Each item has a number or letter (1, 2, 3 or a, b, c). Used when items must be in a fixed order, like steps in a method.
2. **Bulleted List:** Each item has a bullet (a dot or symbol) with no number. Used when the order of items doesn't matter.

Both make steps and points easy to read. A bullet is usually a black circle but can be another symbol.

**To create a bulleted list:**

1. Click the **Bullets** button.
2. Type the first item and press **Enter**.
3. The next line gets a new bullet automatically.
4. Type the next item and press Enter.
5. When the list is done, press **Enter twice** to stop the list.

**To create a numbered list:**

1. Click the **Numbering** button.
2. Type the first item and press **Enter**.
3. The next line gets the next number automatically.
4. Type the next item and press Enter.
5. When the list is done, press **Enter twice** to stop the list.

**Difference between Save and Save As (2.1.11)**

1. **Save:** Stores new work in the same file and folder, updating the last saved version. Used when the name or place of the document should **not** change.
   - **Shortcut:** Ctrl + S (also Shift + F12 or Alt + Shift + F2).
2. **Save As:** Saves the file for the first time, or lets us change its name, place or **format**, while keeping the original file safe as a backup.
   - **Shortcut:** F12 (or Alt + F + A).

**Save vs Save As**

| Save | Save As |
|---|---|
| Stores the data back in the same file and folder | Stores a new file, or the file in a new place |
| Prevents data loss and keeps the file updated with the latest work | Saves an existing file in a new place, with the same or a different name |
| Works on the current file | Can create a new file |
| Has only one step | Needs some extra steps |
| Does not let us save in another file format | Lets us change the file format |

**Insert and format a table (2.1.12)**

1. Place the cursor where you want the table.
2. Click the **Insert** tab.
3. Click **Table** in the Tables group.
4. Move the pointer across and down the grid, and pick how many rows and columns you want.
5. The selected cells turn orange; click, or press **Enter**, to insert the table.

**Header, footer and page numbers (2.1.13)**

Headers and footers add extra information, like the title, file name, date and page numbers, so a document looks neat and easy to read.

1. **Header:** Appears in the top margin of every page.
2. **Footer:** Appears in the bottom margin of every page.

**Steps:** Open the document, click the **Insert** tab, then click Header, Footer or Page Number in the Header & Footer group.

**Long Question: Elaborate on the use of thesaurus and synonyms features in Microsoft Word.** *[Exercise — Brief Q5]*

The thesaurus in MS Word finds synonyms and antonyms for a word, so we can pick a better word instead of repeating the same one.

1. **Thesaurus:** A tool in MS Word (2.1.14) that finds **synonyms** (same meaning) and **antonyms** (opposite meaning) for a chosen word.
2. **Use:** Used mainly when writing an email, letter, project report or research paper, to pick a better word and avoid repeating the same one.
3. **Steps to use synonyms:**
   - Open a new Word document, or an existing one.
   - Type a word, or find a word already in the document.
   - Select (highlight) the word. A blue background appears behind it.
   - Right-click on the word, and a menu appears.
   - Move the cursor to **Synonyms**, then click the best word from the list.
4. We can also open the full Thesaurus box from the Synonyms menu, or from the **Review** tab.

**Apply a procedure to print a document (2.1.15)**

A document can be printed easily if a printer is connected to the computer and its software is installed.

1. Open the Word document — double-click it, or go to **File > Open**.
2. Go to **File > Print**.
3. Choose the print options — number of copies, printer, pages — then click **Print** or **OK**.

## TOPIC 5: MULTIMEDIA PRESENTATION (POWERPOINT)

**Create a multimedia presentation (2.2)**

A **multimedia presentation** is a set of media objects, such as pictures, text, video clips and sound, shown one after another on slides, at a fixed time and place on the screen.

The easiest way to make a multimedia presentation is **Microsoft PowerPoint**, where we can add video, sound and animation. Its biggest advantage is that anyone can edit the presentation easily.

**Adding new slides (2.2.1)**

1. On the **Home** tab, click **New Slide**.
2. In the New Slide box, pick the layout you want — for example, Title Slide, Title and Content, Section Header.
3. Click **Add Slide**.

**Adding various objects on the slides (2.2.2)**

PowerPoint lets us add clip art, images, audio and graphics, and bring in files in different formats. It also has its own gallery of clip art.

**To insert clip art:** Choose a slide layout that has a content box (placeholder), then click the **Insert Clip Art** button inside it.

**To insert an object:**

1. Click on the slide where you want to place the object.
2. On the **Insert** tab, in the Text group, click **Object**.
3. If the object doesn't exist yet, click **Create New**.
4. To show the content on the slide, clear (untick) the **Display as icon** check box.

### Adding notes (2.2.3)

**Speaker notes** help the speaker remember and stress the main points during a talk. The audience does **not** see them.

1. Open the presentation, then click **View > Notes**, or click the **Notes** button at the bottom of PowerPoint.
2. Click “Click to add notes” and type notes for that slide. There is no limit to how many can be added.

### Creating a slide show and an executable file (2.2.4)

Once all slides are ready, press **F5** to start the show from the first slide, and use the arrow keys to move between slides.

**To set up the slide show:** Go to **Slide Show > Set Up Slide Show** and pick one option:

1. **Browsed by an individual (window):** The show runs inside a window. Works best when saving the show on a CD.
2. **Browsed at a kiosk (full screen):** The show runs by itself, full screen. Works best when turning the show into a video.

**Add timings to each slide:** Go to **Slide Show > Rehearse Timings**, and the show opens in full screen with a Recording toolbar and a timer.

- **Next:** go to the next slide.
- **Pause:** stop the timer.
- **Repeat:** record the time again for that slide.

At the end, close the toolbar and click **Yes** to keep the timings.

**Edit timings of a slide:** Go to **View > Slide Sorter**, click **Transitions**, pick a slide, then type the time in the **Advance Slide After** box and press **Enter**.

### Explain the importance of a presentation tool (2.2.5)

**Long Question: Write a note on the benefits of Microsoft PowerPoint, which add importance to a presentation for any kind of user.** *[Exercise — Brief Q1]*

Microsoft PowerPoint is one of the most popular presentation tools, used by people of all ages. It shows ideas in an attractive way, and makes business meetings more interesting than plain talking.

Its main benefits are:

1. **Point to Point Focus:** It keeps the presentation focused on main points, using short points with pictures, sound or animation instead of long paragraphs, saving time and effort.
2. **Attractive Visuals:** It gives the slides a striking look, with colourful backgrounds, text and animation — pictures stay in our mind longer than spoken words.
3. **Numerous Resources:** It gives us many features, and material or references from the Internet can be added with a single click.
4. **Breaking the Complexity:** It helps the presenter explain hard ideas in a simple way, using a laser pointer to point at the bullet points while keeping eye contact with the audience.
5. **User Ownership:** It lets us add a digital signature, so nobody can change the presentation without the owner’s permission.
6. **Multimedia Capability:** It lets a presentation be turned into a video or audio file, shared on the Internet, saved on a DVD, or viewed on mobile phones and other devices.
7. **Easy to Access:** It lets us send slides by email easily, or share them on storage devices, like a USB drive.

**Long Question: Prepare a presentation on the book the students read last time, using all the tools.** *[Exercise — Detail Q1]*

We can make a presentation in Microsoft PowerPoint with these steps:

1. **Open PowerPoint:** Open Microsoft PowerPoint and choose a Blank Presentation.

2. **Title Slide:** On the first slide, type the name of the book and your own name.
3. **Add New Slides:** Click **Home > New Slide**, pick a layout like Title and Content, and add slides for the author, the story, the main characters and the lesson of the book.
4. **Add Text:** Click inside the placeholder and type short points about the book.
5. **Add Pictures, Audio and Video:** Click the **Insert** tab, and use Pictures, Audio or Video to add the book cover or other media.
6. **Add Speaker Notes:** Click **View > Notes**, and write what you will say for each slide.
7. **Add Timings:** Use **Slide Show > Rehearse Timings** to set a time for each slide.
8. **Start the Slide Show:** Press **F5** to show the presentation from the first slide, using arrow keys to move between slides.
9. **Save:** Press **Ctrl + S** to save the presentation. It can also be saved as a video to share with others.

## TOPIC 6: ELECTRONIC MAIL (E-MAIL)

**Electronic mail**, or **email**, is a way to communicate using electronic devices to send messages across networks.

### About email (2.3)

**Gmail** (Google mail) works like an email client. It connects to the Gmail server, and also to other mail servers using the **POP3** or **IMAP** protocol (a protocol is a set of rules). Gmail shows the mailbox in many panes (parts), so we can read mail easily, and it also has a junk mail filter that blocks unwanted mail.

### Create an electronic mail — things we need (2.3.1)

To send an email, we need:

1. An email address.
2. A password.
3. The email address of the person we want to send the email to.
4. An Internet connection.

### Important terms

1. **Email Address:** The unique address of an email account, used to send and receive mail. Example: `testclass7@gmail.com`, made of a user name, the @ sign and the domain name.
2. **Password:** A secret word made of letters, numbers and symbols that protects our account so no one else can open it.
3. **Recipient:** The person who receives the email — their address goes in the To, Cc or Bcc box.
4. **Attachment:** A computer file that is sent along with an email message.
5. **Contacts:** A saved list of names and email addresses of people we talk to.

### Types of emails we should remove from a contact list

| Type | Meaning |
|---|---|
| Bounced | An email that cannot be delivered, or has a delivery problem |
| Malformed | An email address in the wrong format; it does not meet the rules |
| Fake | A fake address made by a fake email generator |
| Inactive | An address that gets newsletters but the owner never reads them |

| Type | Meaning |
|---|---|
| Service | An address like `support@company.com` or `team@company.com` |
| Unsubscribed | An address whose owner has refused to get our messages |
| Duplicate | An email address written in the list more than once |
| Spam Trap | An address made on purpose to catch spammers |

**Use an email service to create an email account (2.3.2)**

1. Click the **Compose** button in the left panel.
2. Enter the recipient's address.
   - **To:** Has the address of the main (primary) recipient.
   - **Cc (Carbon Copy):** Sends a copy to another (secondary) address; everyone can see who got a Cc.
   - **Bcc (Blind Carbon Copy):** Sends a hidden copy to someone; the other recipients don't know about it.
   - If the recipient is in your Address Book, click To, Cc or Bcc to open the Select Names box.
   - To send an email to more than one person, put a **semicolon (;)** between the addresses.
3. Type the subject in the Subject line.
4. Type your message in the body of the email.
5. Click **Send**.

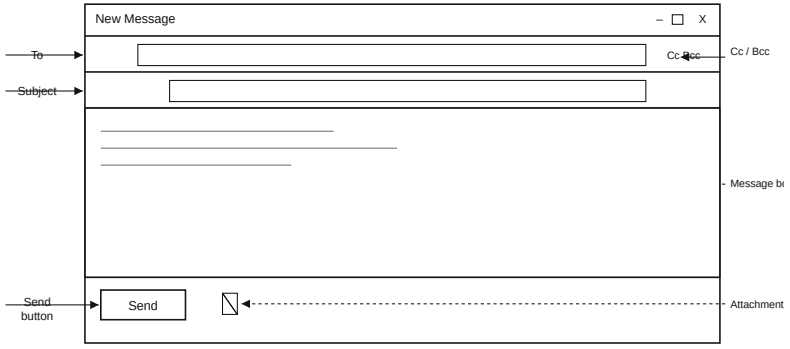



*Fig: Parts of a new email message window — To, Cc/Bcc, Subject, message body, attachment and Send button*

**Send and receive emails with attachments (2.3.3)**

An **attachment** is a computer file sent inside an email message, usually seen in the Reading Pane, under the subject or in the message body. Double-click the attachment icon to open it.

**Files we can attach:**

| File type | Full name |
|---|---|
| bmp | Bitmap Image File |

| File type | Full name |
|---|---|
| gif | Graphical Interchange Format File |
| jpg / jpeg | JPEG Image |
| pdf | Portable Document Format File |
| doc / docx | Microsoft Word Document |
| xls / xlsx | Microsoft Excel Spreadsheet |
| csv | Comma-Separated Values |
| zip | Data Compression File |

**Good habits for writing an email**

The tone and language of an email should be polite, clear and correct.

1. Use a clear, professional subject line.
2. Proofread (check) every email before you send it.
3. Write the email first, then type the recipient's address.
4. Double-check that you have the correct recipient.
5. Make sure you Cc everyone who needs it.
6. You don't always have to click "Reply all".
7. Reply to your emails.
8. Add a signature block at the end.
9. Use the right level of formality — formal or informal.
10. Keep emails short and to the point.

**Organize emails using folders and labels (2.3.4)**

Folders help us arrange email messages, calendars, contacts and tasks. In Gmail, folders are called **labels**.

**To add a folder:** In the left pane, right-click where you want it, then click **New Folder**. The new folder can then be renamed.

**Log in to an email inbox (2.3.5)**

1. Go to the `mail.google.com` page.
2. Click the Gmail **Sign in** option.
3. Enter the email address and click **Next**.
4. Enter the password and click **Next**.

**An appropriate use of email — signing out (2.3.6)**

**Long Question: Write the proper way of signing out of an email account when not using it.** *[Exercise — Brief Q4]*

After using email, we should always sign out of our account, especially on a computer that is not our own.

1. **Right use:** After using email, we should always sign out (log out) of our account. This matters most on a computer that is not our own, such as a school lab or cafe computer.
2. **Reason:** If we don't sign out, another person can open our account, read our emails, or send wrong emails in our name.

3. **Steps to sign out of Gmail:**
   - Click the user name (profile picture) at the top of the Gmail page — a menu of options appears.
   - From these options, click **Sign out**.
4. **Other good habits:** Never share your password, and don't let the browser save your password on a shared computer.

## TOPIC 7: EMAIL AUTHENTICATION, USES, PLATFORMS AND PROTOCOLS

### Use email as a means of authentication for another website login (2.3.7)

**Long Question: State the different ways in which we can use email as a means of authentication for another website.** *[Exercise — Brief Q2]*

Many websites use our email address to check who we are. When we make an account or log in elsewhere, it sends a link or a code to our email to prove the email is ours.

**Three common authentication types:**

1. **Password-based Authentication:** Uses an email address and a password — the most common method.
2. **Certificate-based Authentication:** Uses a digital certificate (a digital ID card) to identify a user or a device.
3. **Biometric Authentication:** Uses body features, like a fingerprint or a face.

(The book's summary also names two more types: **Multi-factor authentication**, where more than one check is used — like a password plus a code — and **Token-based authentication**.)

**Three major methods used in email authentication**, all based on DNS TXT records:

1. **DKIM (DomainKeys Identified Mail):** The sending mail server adds a secret signature to the email with a **private key**, and the receiving server checks this signature with a **public key** kept in the DNS.
2. **SPF (Sender Policy Framework):** The receiving mail server checks the DNS to see if the sender is allowed to send emails, then the email goes to the inbox, junk folder, is quarantined (held back), or blocked.
3. **DMARC (Domain-based Message Authentication, Reporting and Conformance):** It uses both SPF and DKIM. If the email passes, it goes to the inbox; if it fails, it is rejected or sent to the spam folder (quarantine). A report is sent back to the sender.

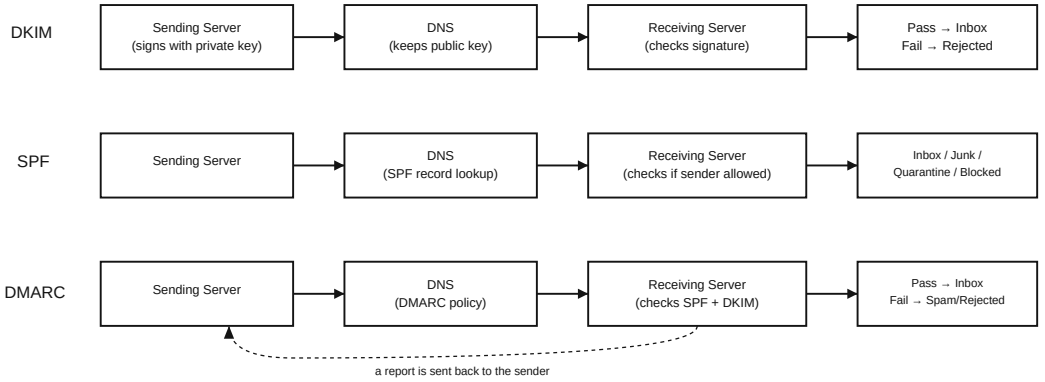



*Fig: How DKIM, SPF and DMARC authenticate an email*

### Uses of electronic mail (2.3.8)

**Long Question: Discuss the importance and uses of email.** *[Exercise — Brief Q3]*

**Importance:** Today, people want things fast and want to save time. Email is quick, cheap and easy, which is why it is used and recommended worldwide.

**Uses of email:**

1. We can talk to people all over the world for free.
2. We can contact more than one person at a time with group mails.
3. We can keep a record of our documents and talks (conversations).
4. We can work together with others (collaboration).
5. We can send files as attachments.
6. We can keep one conversation going for many people together.
7. We can make calendars and set appointments.

### The common platforms for electronic mail (2.3.9)

**Long Question: Identify and explain the common platforms for electronic mail.** *[Exercise — Brief Q8]*

An **email platform** is a website or service that gives us an email account. Some common platforms are:

1. **Gmail:** The free email service from Google, and the most used platform.
2. **Microsoft Outlook:** The email service from Microsoft (Outlook.com).
3. **Yahoo Mail:** The free email service from Yahoo.
4. **AOL Mail (AOL.com):** An old and well-known free email service.
5. **Zoho Mail:** An email service mostly used by businesses.
6. **ProtonMail:** An email service known for strong security and privacy.

With these platforms, we can send, receive and organise emails from any computer or phone with an Internet connection.

### Create an e-mail address, and how to send electronic mail (2.3.10)

**Long Question: Create an email address using common, freely available platforms like Microsoft or Google.** *[Exercise — Detail Q2]*

With a free Gmail account, we can use our email, calendar, tasks and contacts from anywhere with an Internet connection.

**Steps to create a Gmail account:**

1. Open a web browser, go to the `gmail.com` sign-up screen, and click **Create account**.
2. Choose “For myself” — a new window opens.
3. Enter your first name, last name and a user name.
4. Enter a password and confirm it, then press **Next** (use 8 or more characters, with a mix of letters, numbers and symbols).
5. Enter a phone number and a recovery email address — these are optional (not compulsory).
6. Enter your birthday.
7. Choose your gender from the drop-down list and click **Next**.
8. Read the privacy policy and terms, then click **“I agree”**.
9. Gmail sets up your account and shows a welcome screen — you can now use the account on computers and mobile phones.

**Process of authentication (2.3.11)**

**Email authentication** is a technical way to check that an email is not forged (fake); it proves the email comes from the person it claims to be from. It is mainly used to stop harmful and fraudulent emails, such as spam.

**Email protocols (2.3.12)**

**Long Question: Explain the standard protocols SMTP and POP3.** *[Exercise — Brief Q10]*

An **email protocol** is a set of rules that lets emails move between servers and email programs (clients) in a standard way, so email works the same for every user everywhere.

The common protocols are:

1. **POP (Post Office Protocol):** The protocol email clients (programs) use to download emails from the server. It's mostly one-way, and does **not** sync (copy) changes back to the server. **POP3** is its version 3.
2. **IMAP (Internet Message Access Protocol):** The standard protocol for incoming email. It keeps emails on the mail server, so the user can see and manage them as if they were saved on their own device, and changes stay the same on every device.
3. **SMTP (Simple Mail Transfer Protocol):** The main protocol for **sending** email — it moves emails between email clients and email servers.

SMTP sends the email, while POP3 or IMAP brings it to the receiver.

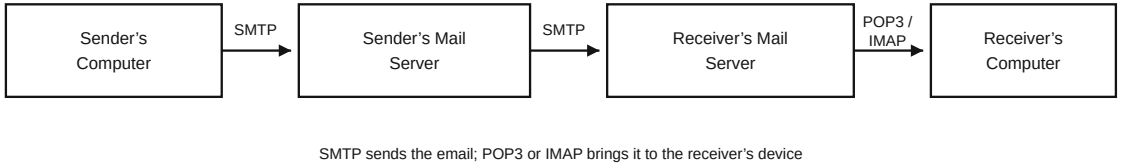



*Fig: How SMTP, POP3 and IMAP move an email from sender to receiver*

## SHORT QUESTIONS — UNIT 2

1. **Describe the purpose of a word processor.** *[Topic 1 · Exercise]*
   A word processor is used to type, edit, format, save and print text. It helps us make letters, reports, tests and homework.
2. **Write the four functions of a word processor.** *[Topic 1 · Book concept]*
   The four functions are composing, editing, saving and printing.
3. **Give any four examples of word processors.** *[Topic 1 · Book concept]*
   MS Word, Google Docs, WordPerfect and LibreOffice Writer.
4. **Write a note on the benefits of Microsoft PowerPoint.** *[Topic 5 · Exercise]*
   PowerPoint keeps a talk focused on the main points. It gives attractive visuals. It adds resources from the Internet. It explains hard ideas simply. It allows a digital signature. It can be turned into a video. It is easy to share by email.

5. **State the ways we can use email for authentication on another website.** *[Topic 7 · Exercise]*
   Websites send a link or a code to our email to prove who we are. Common types are password-based, certificate-based and biometric authentication. Email itself is checked with DKIM, SPF and DMARC.
6. **Discuss the importance and uses of email.** *[Topic 7 · Exercise]*
   Email is fast, cheap and easy. We use it to talk to people all over the world for free. We can send group mails, send attachments, work together, and make calendars and appointments.
7. **Write the proper protocol for signing out of an email account.** *[Topic 6 · Exercise]*
   Click the user name at the top of the Gmail page. Then click Sign out. We should always sign out on a computer that is not our own.
8. **Elaborate on the use of thesaurus and synonyms in MS Word.** *[Topic 4 · Exercise]*
   The thesaurus finds synonyms (same meaning) and antonyms (opposite meaning) for a word. Select the word, right-click, go to Synonyms, and click the best word.
9. **Differentiate between Save and Save As.** *[Topic 4 · Exercise]*
   Save updates the same file with your new work (Ctrl + S). Save As saves the file for the first time, or with a new name, place or format (F12).
10. **Identify the common platforms for electronic mail.** *[Topic 7 · Exercise]*
    Gmail, Microsoft Outlook, Yahoo Mail, AOL Mail, Zoho Mail and ProtonMail.
11. **Define recipient.** *[Topic 6 · Exercise]*
    The person who receives an email is called the recipient.
12. **Define attachment.** *[Topic 6 · Exercise]*
    A computer file that is sent along with an email message is called an attachment.
13. **Define password.** *[Topic 6 · Exercise]*
    A password is a secret word of letters, numbers and symbols that protects our account.
14. **Define email address.** *[Topic 6 · Exercise]*
    It is the one-of-a-kind address of an email account, where emails are sent and received, such as testclass7@gmail.com.
15. **Explain the standard protocols SMTP and POP3.** *[Topic 7 · Exercise]*
    SMTP (Simple Mail Transfer Protocol) sends emails between clients and servers. POP3 (Post Office Protocol) downloads emails from the server to the user's device.
16. **What is the difference between POP3 and IMAP?** *[Topic 7 · Book concept]*
    POP3 downloads emails and does not sync back to the server. IMAP keeps emails on the server, so they look the same on every device.
17. **What is the difference between Cc and Bcc?** *[Topic 6 · Book concept]*
    Cc sends a copy that all recipients can see. Bcc sends a hidden copy that other recipients do not know about.
18. **What is a multimedia presentation?** *[Topic 5 · Book concept]*
    It is a set of media objects, such as text, pictures, video and sound, shown on slides. The easiest way to make it is MS PowerPoint.
19. **What are speaker notes?** *[Topic 5 · Book concept]*
    Speaker notes are notes added to a slide to help the presenter remember the main points. The audience does not see them.

## MULTIPLE CHOICE QUESTIONS — UNIT 2

These are the book's own Exercise MCQs plus a few extra practice MCQs, with the correct answer marked ✔ above the option.

1. To open a word document, we go to the ____ tab. *[Topic 1 · Exercise MCQ 1]*

   a) Open  **b) File** ✔  c) Home  d) Insert

2. ____ is a word processor that allows us to enter, format, save and print text. *[Topic 1 · Exercise MCQ 2]*

   a) MS Excel  b) MS PowerPoint  **c) MS Word** ✔  d) MS Paint

3. We can format the text from the ____ tab. *[Topic 2 · Exercise MCQ 3]*

   a) File  b) Format  c) Insert  **d) Home** ✔

4. ____ makes text appear thicker and darker. *[Topic 2 · Exercise MCQ 4]*

   a) Italics  b) Format  c) Font face  **d) Bold** ✔

5. There are ____ types of alignment in text formatting in Word. *[Topic 2 · Exercise MCQ 5]*

   a) 3  **b) 4** ✔  c) 5  d) 6

6. The ____ tab lets you control the look and feel of your document in MS Word. *[Topic 2 · Exercise MCQ 6]*

   **a) Layout** ✔  b) Margin  c) Caption  d) Format

7. We can insert an image in the Word document from the ____ tab. *[Topic 3 · Exercise MCQ 7]*

   **a) Insert** ✔  b) File  c) Home  d) View

8. We can resize an image by clicking and dragging on its ____. *[Topic 3 · Exercise MCQ 8]*

   a) Outline  b) Center  **c) Resize handles** ✔  d) Arrows

9. The keyboard shortcut to copy is ____. *[Topic 3 · Exercise MCQ 9]*

   a) Ctrl + X  **b) Ctrl + C** ✔  c) Ctrl + P  d) Ctrl + S

10. We can insert a table in the Word document from the ____ tab. *[Topic 4 · Exercise MCQ 10]*

    a) File  b) Home  **c) Insert** ✔  d) View

11. ____ appears at the top margin of the Word document. *[Topic 4 · Exercise MCQ 11]*

    a) Footer  b) Title  c) Address bar  **d) Header** ✔

12. To print the Word document, we go to the ____ tab. *[Topic 4 · Exercise MCQ 12]*

    a) Print  **b) File** ✔  c) Format  d) Insert

13. The easiest way to create a multimedia presentation is on ____. *[Topic 5 · Exercise MCQ 13]*

    a) Microsoft Word  b) Microsoft Excel  **c) Microsoft PowerPoint** ✔  d) Microsoft Paint

14. The keyboard shortcut to create a new email is ____. *[Topic 6 · Exercise MCQ 14]*

    a) Ctrl + S  b) Ctrl + N  **c) Ctrl + M** ✔  d) Ctrl + X

15. It is always safe to ____ your account, if you are not using it. *[Topic 6 · Exercise MCQ 15]*

    a) Protect  **b) Sign out** ✔  c) Sign in  d) Delete

16. How many primary functions does a word processor have? *[Topic 1]*

    a) two  b) three  **c) four** ✔  d) five

17. The keyboard shortcut to cut is: *[Topic 3]*

    a) Ctrl + C  **b) Ctrl + X** ✔  c) Ctrl + V  d) Ctrl + Z

18. The keyboard shortcut to paste is: *[Topic 3]*

    a) Ctrl + P  b) Ctrl + C  **c) Ctrl + V** ✔  d) Ctrl + X

19. Cut or copied text is kept in the: *[Topic 3]*

    a) Recycle Bin  **b) Clipboard** ✔  c) Ribbon  d) Title Bar

## Mistakes Found in the Book

1. **Alignment count (page 20, Exercise MCQ 5):** The book's own answer key gives (a) 3 as the correct answer. But the book's own body text, on page 20, shows and names **four** types of alignment: Left Align, Center Align, Right Align and Justify. So the correct answer should be (b) 4. These notes use the book's own body text and mark (b) 4 as correct.
2. **New email shortcut key (Exercise MCQ 14, and the Summary on page 38):** The official answer key says the answer is (c) Ctrl + M. But the book's own **Summary** box, at the end of the unit, states: *"Click the New Email button on the Home tab of Mail. Press Ctrl + N."* The book contradicts itself: one part says Ctrl + N, the answer key says Ctrl + M. In real Microsoft Outlook, a new email is made with Ctrl + N (or Ctrl + Shift + M). Since this is a graded exam answer, these notes tick (c) Ctrl + M to match the official answer key. Students should know the book itself disagrees with its own key.
3. **Number of authentication types (page 34 vs page 39):** Page 34 (section 2.3.7) says there are **3** common authentication types: password-based, certificate-based and biometric. But the Summary on page 39 lists **5** types: it adds multi-factor authentication and token-based authentication. These notes give the 3 main types in full, and mention the other 2 so nothing is missed.
4. **Wrong app name in a heading (page 33, section 2.3.5):** The heading says "Log in an email inbox," and the paragraph below it talks about signing in to **Outlook**. But every step given is for **Gmail** (`mail.google.com`, Gmail Sign in). The book mixed up Outlook and Gmail instructions.
5. **Typo in the uses of email list (page 35, section 2.3.8):** The book's list of email uses has the line "**rating** calendars and appointments." This is a typo for "**creating** calendars and appointments," which fits the rest of the sentence.
6. **"Compose email using Word" (page 30, section 2.3):** The book's intro to email says, "You can use Word to compose email." This line does not fit with Gmail, which the rest of the section is about: Gmail is used through a web browser, not Microsoft Word. It looks copied from different source material without being updated.
7. **Heading promises something the text never explains (page 28, section 2.2.4):** The heading is "Creating a SlideShow **and an executable file**." But nowhere in this section does the book explain how to turn a presentation into an executable (.exe) file. It only explains how to start a slide show (F5) and set slide timings. Students should know the book does not teach "executable file" creation, even though the heading mentions it.

# Unit 3 — Computational Thinking

## TOPIC 1: COMPUTATIONAL THINKING

### What is Computational Thinking? (3.1)

Computational Thinking (CT) is a set of skills used to solve problems. It helps us to:

1. Understand a complex problem.
2. Find possible solutions.
3. Show solutions in a way that a computer or human can understand.

**Venn diagram (Fig. 3.1):** Computing is the larger field. Computational Thinking and Computer Science are both parts of it, and they overlap with each other.

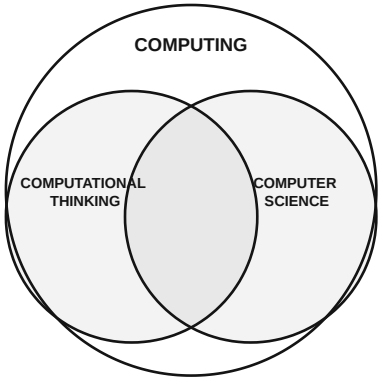



*Fig: Venn diagram of Computational Thinking — Computing is the big circle, with Computational Thinking and Computer Science as two overlapping circles inside it*

### Cornerstones (Main Techniques) of Computational Thinking

Computational thinking rests on four key techniques:

1. **Decomposition:** Breaking a task or problem into small steps or parts.
2. **Pattern Recognition:** Finding things that are similar between problems.
3. **Generalization and Abstraction:** Finding the main rules (principles) that cause these patterns.
4. **Algorithm Design:** Making step-by-step instructions that solve a problem and can be reused for similar problems.

**Extra Bit:** Interdisciplinary studies means joining two or more subjects into one activity.

### Decomposition

Decomposition means breaking a big, complex problem into smaller parts, since smaller problems are easier to solve. It is an important life skill.

**Example:** Hosting a holiday dinner can be broken into small parts:

1. Select the menu.
2. Get help from others in the kitchen.

3. Tell people what to bring.
4. Decide how each dish will be cooked.
5. Set the time for the event.

**Diagram:** Decomposition can be drawn as a tree: a Big Problem splits into Smaller Problems, and each of those splits again into Small Parts.

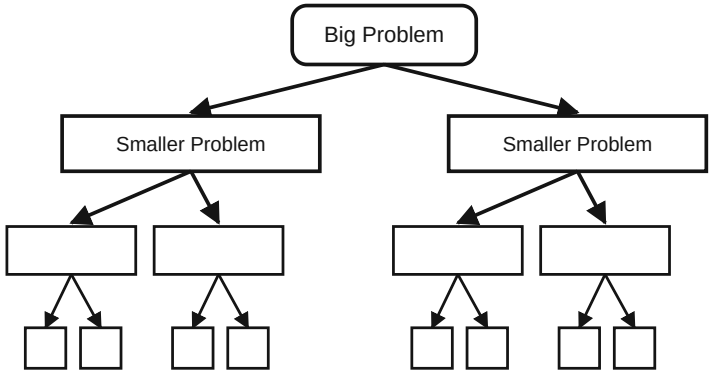



*Fig: Decomposition tree — a Big Problem splits into Smaller Problems, which split further into Small Parts*

**Pattern Recognition**

Pattern recognition means looking for patterns in a problem. A pattern is something that repeats or looks similar. It checks whether a problem or solution seen before can be reused, including similar shapes, sounds, or pictures.

**Example:** In a computer game, we look for similar objects, patterns, and actions. Using the same method, or a slightly changed one, for all of them makes the solution faster and more efficient.

**Do you know?** Patterns are the laws of nature and life, found in every subject.

**Generalization and Abstraction**

1. **Abstraction:** It means picking out the details that matter for solving the problem, and ignoring the ones that don't.
2. **Generalization:** It means forming a general idea about the problem and its solution. Patterns found are written in their simplest form, for reuse whenever needed.

**Example:** A car and a bicycle are wheeled vehicles. A boat is a vehicle too, but it has no wheels. Abstraction keeps the main idea — all three are vehicles — and ignores small details like colour and size.

**Long Question: Define pattern recognition in your words. [Exercise, Detailed Q1]**

Pattern recognition means looking for patterns, or similar things, in a problem, so they can be solved the same way.

1. **Meaning:** It means looking for patterns, or similar things, in a problem.
2. **What we check:** Whether a problem or solution solved before can be reused, including similar shapes, sounds, or pictures.
3. **Use:** Finding similarities between the small problems we get after decomposition.
4. **Example:** In a computer game, we find similar objects, patterns, and actions, then use the same method, or a slightly changed one, for all of them. This makes the solution faster and better.

## TOPIC 2: ALGORITHMS

### Algorithm Design

An algorithm is a set of instructions that solves a problem or completes a task: a list of steps a human or computer can follow to get an answer or solution, in a limited (finite) number of steps.

### Characteristics of an Algorithm

1. **Clear and Unambiguous:** The algorithm must be clear, with every step having only one possible meaning.
2. **Well-Defined Inputs:** The algorithm's inputs must be clearly stated.
3. **Well-Defined Outputs:** The algorithm must clearly say what the output will be.
4. **Finite-ness:** The algorithm must end (stop) after a limited time.
5. **Feasible:** The algorithm must be simple and practical, doable with the resources we have.
6. **Language Independent:** The algorithm must not depend on any one programming language.

**Diagram (Fig. 3.1):** An algorithm has six characteristics: Well-Defined Inputs, Well-Defined Outputs, Clear and Unambiguous, Finite-ness, Language Independent, and Feasible.

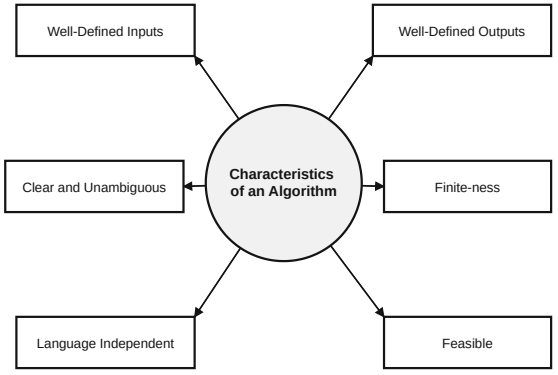



*Fig: Characteristics of an algorithm — six characteristics joined to the central idea by arrows*

### Types of Algorithms

1. **Brute Force Algorithm:** It is the simplest method, trying every possible solution, one by one, until it finds the right one.
2. **Recursive Algorithm:** It breaks the problem into smaller sub-parts, and calls the same procedure (method) again and again to solve them.
3. **Searching Algorithm:** It searches for one item, or a group of items, inside given data.
4. **Sorting Algorithm:** It arranges a group of data in a certain order, e.g., smallest to largest.
5. **Divide and Conquer Algorithm:** It breaks a problem into sub-problems, solves one of them, then merges the solutions to get the final answer.
6. **Randomized Algorithm:** It uses a random number to give an immediate benefit, helping decide the expected result.

### Structures of an Algorithm

1. **Sequence Structure:** The steps are carried out in the exact order they are written.
2. **Selection (Conditional) Structure:** A step, or group of steps, is carried out only if a condition is true; otherwise the "else" part runs. **Example:** An "if-then-else" condition checks if a number is even or odd.

3. **Repetition (Loop) Structure:** A loop is a group of instructions carried out again and again, as long as a condition stays true. Once the condition becomes false, the loop stops, and the steps after it are carried out.

**Types of Loops**

1. **Finite Loop:** It has a clear end, running a fixed number of times and stopping once the condition becomes false. It is the most common type of loop.
2. **Infinite Loop:** It has no clear end, and runs forever because its condition always stays true. **Example:** The “forever” block in Scratch makes a sprite repeat its action forever.

**Computational Thinking in Real Life**

Computational thinking can be seen in how people make decisions, do simple maths, and solve everyday problems:

1. Using an algorithm to find the best route from school to home, thinking about traffic, construction work, or road blocks.
2. Following a recipe to bake a cake, itself an example of an algorithm.
3. Planning a budget, using pattern recognition and decomposition to check spending in each category.

**Essentials (Pre-requisites) of Writing an Algorithm**

Before we write an algorithm, there are five things we need to know:

1. **The Problem:** A clear definition of the problem we need to solve.
2. **The Limitations (Constraints):** The limits we must keep in mind while solving it.
3. **The Input:** The data we take in to solve the problem.
4. **The Output:** The result we expect once the problem is solved.
5. **The Process (Solution):** The method we use to solve it, staying inside the given limits.

Once we know all five, we’re ready to write the algorithm.

**Long Question: Write an algorithm to find the sum of three numbers. [Exercise, Detailed Q2]**

To find the sum of three numbers, we first list the pre-requisites, then write the algorithm step by step.

**Step 1: Pre-requisites**

1. Problem: We need to add three numbers and print their sum.
2. Constraints: The numbers must contain only digits — no other characters are allowed.
3. Input: The three numbers we want to add.
4. Output: The sum of the three numbers (a single integer value).
5. Solution: We add the three numbers using the ‘+’ operator.

**Step 2: Algorithm — Sum of three numbers**

1. START
2. Declare 3 integer variables: num1, num2 and num3.
3. Take the three numbers as input in num1, num2 and num3.
4. Declare an integer variable called sum.
5. sum = num1 + num2 + num3
6. Print the value of sum.
7. END

**Long Question: Write the algorithm to find whether a given number is even or odd. [Exercise, Detailed Q3]**

To check whether a number is even or odd, we first list the pre-requisites, then write the algorithm step by step.

**Step 1: Pre-requisites**

1. Problem: We need to print whether a number is even or odd.
2. Constraints: The number must contain only digits — no other characters are allowed.
3. Input: A number that needs to be checked.
4. Output: We print “even” or “odd”.
5. Solution: We divide the number by 2 and take the remainder — if the remainder is zero, the number is even, and if not, it is odd.

**Step 2: Algorithm — Even or odd**

1. START
2. Declare an integer variable num1.
3. Take the number as input in num1.
4. Declare an integer variable called remainder.
5. remainder = num1 % 2 (the % sign gives the remainder)
6. IF remainder = 0, print “Number is Even”. Otherwise, print “Number is Odd”.
7. END

**Extra Bit:** “Mod” (%) is used to find the remainder (modulus) of a division. **Example:** 7 % 2 = 1 and 8 % 2 = 0.

## TOPIC 3: FLOWCHARTS

### What is a Flowchart?

A flowchart is a picture (symbolic) form of an algorithm, where each step is shown by a symbol with a short description inside it. Arrows join the steps and show the flow’s direction. Flowcharts are used to study, design, document, and manage a process or a program.

### Commonly Used Symbols in Flowcharts

| Symbol | Name | Use |
|---|---|---|
| Oval | Terminal / Terminator | Shows the start or stop point of a flowchart |
| Rectangle | Action / Process | Shows a process (piece of work) in the flowchart |
| Arrow | Arrow / Flow line | Shows the direction of flow from one step to another |
| Parallelogram | Input / Output | Shows input or output |
| Diamond | Decision | Shows a decision (Yes or No) |

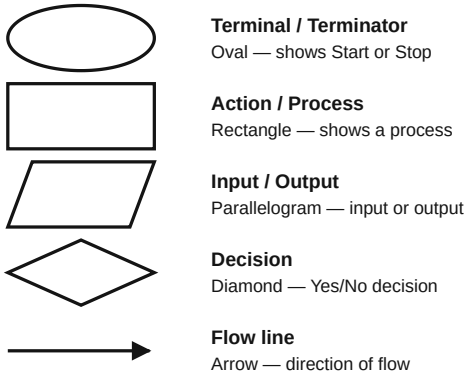



*Fig: The commonly used flowchart symbols*

**Long Question: Write down the rules of writing a flowchart. [Exercise, Detailed Q4]**

A flowchart must follow a few basic rules to stay clear and easy to follow:

1. Use the standard (conventional) flowchart symbols.
2. Every flowchart must have a Start point and an End (Stop) point.
3. Each symbol should have only one exit point, except the decision symbol, which has two (Yes and No).
4. Flow lines should not cross each other.
5. The flow lines coming out of a decision symbol should be labelled properly (Yes / No).

**Advantages and Disadvantages of Flowcharts**

| Advantages | Disadvantages |
|---|---|
| Easy to make | Difficult to show big (complex) programs |
| Easy to understand | Hard to change (modify) |
| Mistakes can be found easily | Takes a lot of time |
| Debugging becomes possible (removing errors is called debugging) | Hard to understand for people who do not know the symbols |
| The logic can be understood easily | |

**Long Question: Draw a flowchart to find the sum of three numbers. [Exercise, Detailed Q6]**

A flowchart can describe an everyday task too, such as polishing shoes:

**Example — Polish your shoes:** Start → Open the shoe to polish → Put polish on brush → Polish one shoe → Put polish on brush → Polish shoe → Stop.

**Answer — Flowchart for sum of three numbers:** This flowchart takes three numbers as input, adds them, and prints the result.

1. **Start** (oval)
2. **Input three numbers A, B and C** (parallelogram)
3. **Sum = A + B + C** (rectangle — this step does the processing)

4. **Print Sum** (parallelogram)
5. **Stop** (oval)

**Long Question: Draw a flowchart to find whether a given number is even or odd. [Exercise, Detailed Q7]**

**Answer:** This flowchart takes a number, finds the remainder when divided by 2, and prints whether it is even or odd.

1. **Start** (oval)
2. **Input a number A** (parallelogram)
3. **Rem = A % 2** (rectangle — find the remainder)
4. **Decision: Is Rem = 0?** (diamond)
   - If **No**, go to “Print Odd” (parallelogram), then go to Stop.
   - If **Yes**, go to “Print Even” (parallelogram), then go to Stop.
5. **Stop** (oval)




*Fig: Flowchart — sum of three numbers*

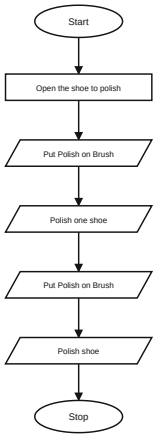



*Fig: Flowchart — Polish your shoe*

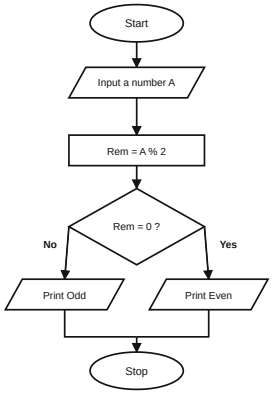



*Fig: Flowchart — even or odd*

**Flowcharts with Loops**

Flowcharts can also use loops. Here are two examples using a finite loop (a loop with a fixed number of repeats):

**Example — Print the first 10 numbers:**

1. Start.
2. Set C = 1.
3. Decision: Is C less than 11?
   - If **No**, go to Stop.
   - If **Yes**, print C. Then set C = C + 1, and go back to the decision step again.
4. Stop.

This loop prints C and adds 1 to it until C is no longer less than 11, printing the numbers 1 to 10.

**Example — Table of a number, up to 10:**

1. Start.
2. Input a number A.
3. Set C = 0.
4. Decision: Is C less than 10?
    - If **No**, go to Stop.
    - If **Yes**, set C = C + 1. Then print “A x C = A*C” (the result of A multiplied by C). Then go back to the decision step again.
5. Stop.

This loop increases C by 1 and prints the multiplication table of A until C is no longer less than 10, building the table one line at a time.

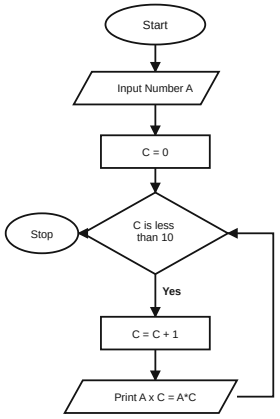



*Fig: Flowchart — table of a number up to 10, a loop*

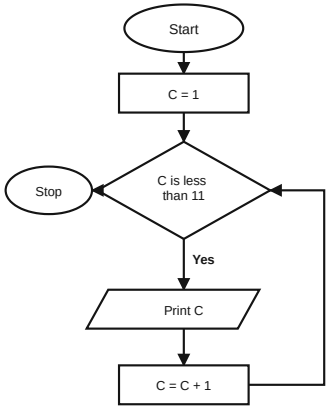



*Fig: Flowchart — print the first 10 numbers, a loop*

## TOPIC 4: EFFICIENCY OF A SOLUTION

One problem can have more than one solution, and some solutions are more efficient (better) than others.

**Long Question: How can we measure the efficiency of a solution? [Exercise, Detailed Q8]**

We measure the efficiency of a solution using two things:

1. **Number of Steps:** How many steps the solution takes to solve the problem.
2. **Memory:** How much memory the solution uses.

But efficiency is mostly measured by the number of steps: the solution with fewer steps is the more efficient one.

**Example — Find the largest number from three numbers:**

**Solution 1 (7 steps, uses 4 variables: A, B, C, largest)**

1. Start.
2. Take three different numbers in A, B and C.
3. Set largest = A.

4. If B is greater than largest, set largest = B.
5. If C is greater than largest, set largest = C.
6. Print largest.
7. Stop.

**Solution 2 (8 steps, uses 3 variables: A, B, C)**

1. Start.
2. Input three different numbers A, B and C.
3. If A is greater than B, go to step 6.
4. If B is greater than C, print “B is largest” and go to step 8.
5. Print “C is largest” and go to step 8.
6. If A is greater than C, print “A is largest” and go to step 8.
7. Print “C is largest”.
8. End.

**Analysis:** Both solutions solve the same problem differently. Solution 1 takes 7 steps and uses four variables (A, B, C and largest); Solution 2 takes 8 steps and uses three variables (A, B and C). Since Solution 1 takes fewer steps, it is more efficient, even though it uses more memory (four variables) than Solution 2 (three).

**Long Question: Why don’t we consider memory when measuring the efficiency of an algorithm? [Exercise, Detailed Q5]**

We don’t consider memory because modern computers have plenty of it; efficiency is measured mainly by the number of steps instead.

1. **Two things to measure:** Efficiency can be measured by the number of steps, and by the memory used.
2. **Memory is not a big problem:** Modern computers have plenty of cheap memory, so it isn’t an issue today.
3. **Steps matter more:** The number of steps decides how fast a problem is solved, so efficiency is always measured by steps.
4. **Example:** In the largest-number example, Solution 1 uses one extra variable (more memory) but takes fewer steps, and is still more efficient.

## SHORT QUESTIONS — UNIT 3

1. **Define computational thinking in your words.** *[Topic 1 · Exercise]*
   Computational thinking is a set of problem-solving skills. It helps us understand a hard problem and make solutions that a computer or a human can understand.
2. **Enlist (list) the techniques of computational thinking.** *[Topic 1 · Exercise]*
   Decomposition, pattern recognition, generalization and abstraction, and algorithm design.
3. **What do you mean by decomposition?** *[Topic 1 · Exercise]*
   Decomposition means breaking a big, hard problem into smaller parts that are easy to solve. Example: dividing the work of a holiday dinner into small tasks.
4. **Elaborate (explain) generalization and abstraction.** *[Topic 1 · Exercise]*
   Abstraction means keeping the important details and ignoring the unimportant ones.
   Generalization means making a general idea of the problem and its solution, so we can use it again.
5. **Define algorithm in your words.** *[Topic 2 · Exercise]*
   An algorithm is a set of step-by-step instructions. It solves a problem, or finishes a task, in a limited number of steps.

6. **Enlist the characteristics of an algorithm.** *[Topic 2 · Exercise]*
   Clear and unambiguous, well-defined inputs, well-defined outputs, finite-ness, feasible, and language independent.
7. **Differentiate between recursive and brute force algorithm.** *[Topic 2 · Exercise]*
   A brute force algorithm tries every possible solution until it finds the right one. A recursive algorithm breaks the problem into sub-parts and calls the same procedure again and again.
8. **Differentiate between searching and sorting algorithm.** *[Topic 2 · Exercise]*
   A searching algorithm finds an item, or a group of items, in given data. A sorting algorithm arranges data in a certain order.
9. **What are loops? Discuss its types.** *[Topic 2 · Exercise]*
   A loop is a set of instructions that repeats again and again while a condition stays true. There are two types: finite loop (stops after a fixed number of times) and infinite loop (never stops).
10. **Enlist the advantages and disadvantages of flowcharts.** *[Topic 3 · Exercise]*
    Advantages: easy to make, easy to understand, mistakes are found easily, debugging is possible, and logic is easy to understand.
    Disadvantages: hard to show big programs, hard to change, and it takes a lot of time.
11. **What is a flowchart?** *[Topic 3 · Book concept]*
    A flowchart is a picture form of an algorithm. Each step is shown by a symbol, and arrows show the flow.
12. **What is pattern recognition?** *[Topic 1 · Book concept]*
    It means finding similar things (patterns) in problems. This lets us use an old solution again.
13. **What is a divide and conquer algorithm?** *[Topic 2 · Book concept]*
    It breaks a problem into sub-problems, solves them, and merges the solutions to get the final solution.
14. **What is a randomized algorithm?** *[Topic 2 · Book concept]*
    It is an algorithm that uses a random number to give an immediate benefit.

## MULTIPLE CHOICE QUESTIONS — UNIT 3

These are the book's own Exercise MCQs plus a few extra practice MCQs, with the correct answer marked ✔ above the option.

1. Breaking down a problem into sub-problems is called: *[Topic 1 · Exercise MCQ 1]*

| a) generalization | b) pattern recognition | **c) deconstruction** ✔ | d) Design |
|---|---|---|---|

2. Discovering the principles that cause the patterns of a problem is called: *[Topic 1 · Exercise MCQ 2]*

| **a) generalization** ✔ | b) pattern recognition | c) deconstruction | d) Design |
|---|---|---|---|

3. A set of instructions to solve a problem is called: *[Topic 2 · Exercise MCQ 3]*

| a) directions | b) instructions | **c) algorithm** ✔ | d) Design |
|---|---|---|---|

4. The algorithm which goes through all possible solutions until the required solution is found is called: *[Topic 2 · Exercise MCQ 4]*

| a) Recursive algorithm | b) Searching algorithm | **c) Brute force algorithm** ✔ | d) sorting algorithm |
|---|---|---|---|

5. The algorithms which help arrange a group of data in a particular manner are called: *[Topic 2 · Exercise MCQ 5]*

| a) Recursive algorithm | b) Searching algorithm | c) Brute force algorithm | **d) sorting algorithm** ✔ |
|---|---|---|---|

6. The algorithm which breaks a problem into sub-problems, solves a single sub-problem, and merges the solutions is called: *[Topic 2 · Exercise MCQ 6]*

| a) Recursive algorithm | b) Brute force algorithm | **c) divide and conquer algorithm** ✔ | d) sorting algorithm |
|---|---|---|---|

7. The algorithm which uses a random number so that it gives an immediate benefit is called: *[Topic 2 · Exercise MCQ 7]*

| a) Recursive algorithm | b) divide and conquer algorithm | c) Brute force algorithm | **d) random algorithm** ✔ |
|---|---|---|---|

8. The sequence where we repeat a specific set of instructions again and again is called: *[Topic 2 · Exercise MCQ 8]*

| **a) loop** ✔ | b) sequence | c) condition | d) all |
|---|---|---|---|

9. The loops which have to be terminated (stopped) are called: *[Topic 2 · Exercise MCQ 9]*

| a) Infinite loops | **b) Finite loops** ✔ | c) intermediate loops | d) simple loops |
|---|---|---|---|

10. The loops which are never going to end are called: *[Topic 2 · Exercise MCQ 10]*

| **a) Infinite loops** ✔ | b) Finite loops | c) intermediate loops | d) simple loops |
|---|---|---|---|

11. CT stands for: *[Topic 1]*

| a) Computer Technology | **b) Computational Thinking** ✔ | c) Common Thinking | d) Coding Technique |
|---|---|---|---|

12. Looking for similar things in problems is called: *[Topic 1]*

| a) decomposition | **b) pattern recognition** ✔ | c) abstraction | d) algorithm design |
|---|---|---|---|

13. Keeping important details and ignoring unimportant ones is called: *[Topic 1]*

| a) decomposition | b) pattern recognition | **c) abstraction** ✔ | d) debugging |
|---|---|---|---|

14. An algorithm must end after a finite time. This characteristic is called: *[Topic 2]*

| a) feasible | **b) finite-ness** ✔ | c) language independent | d) unambiguous |
|---|---|---|---|

## Mistakes Found in the Book

1. **Answer key, Unit 3 MCQ 2:** The book's answer key gives (b) pattern recognition. But the book's own body text (page 44) and Summary call discovering the principles that cause patterns "generalization". The correct answer should be (a) generalization.
2. **Answer key, Unit 3 MCQ 6:** The book's answer key gives (b) brute force algorithm. But the book (page 45) calls this process (breaking into sub-problems, solving, and merging) the divide and conquer algorithm. The correct answer should be (c) divide and conquer algorithm.
3. **Answer key, Unit 3 MCQ 9:** The book's answer key gives (a) infinite loops. But loops that stop (are terminated) are finite loops, not infinite ones. So the correct answer should be (b) finite loops.
4. **Page 54, MCQ 1:** None of the four options is the word "decomposition", even though that is the correct term from the lesson. The closest option is (c) deconstruction, so these notes tick (c) for this question.
5. **Page 55, Detailed Question 1:** The book writes "pattern reorganization" in this question. It should say "pattern recognition".

6. **Page 46:** The book writes “Lepitaton Structure” as a heading. It should say “Repetition Structure”. The same page also misspells “algorithm” as “alogorithm”, and “sequence” as “sequnce”.
7. **Page 46:** The book’s example sentence says “finding whether a member is even or odd”. The word “member” should be “number”.
8. **Page 53, Analysis:** This section says Solution 1 is more efficient “although it consumed less memory”. This is incorrect: Solution 1 uses four variables (A, B, C, largest), which is *more* memory than Solution 2’s three (A, B, C).
9. **Duplicate figure number — “Fig. 3.1”:** The book labels two different diagrams as “Fig. 3.1”. Page 43’s Venn diagram of computational thinking is captioned “Fig. 3.1 Venn Diagram of Computational Thinking”, and page 46’s diagram of the six characteristics of an algorithm is also captioned “Fig. 3.1 Characteristics of an algorithm”. One of the two should have been numbered “Fig. 3.2” instead.

# Unit 4 — Programming

## TOPIC 1: Computers Store Information Using Binary Codes

Computers show everything as numbers, using only two digits: 0 (zero) and 1 (one).

**Binary codes (4.1)**

1. **Binary numbers:** They are numbers written using only 0s and 1s.
2. **Switches:** Computers have millions of tiny electronic switches. An open one means 0; a closed one means 1. This lets a computer show almost anything as numbers.
3. **Bits:** Each switch is a **bit**, short for “Binary Digit”. Bits are what get stored in the computer’s memory.
4. **Byte:** Eight bits together make one **byte** of data.
5. **Memory:** A megabyte holds about a million bytes, a gigabyte about a billion. The more memory a computer has, the more information it can store.

**Do you know?** Bit is the short form of Binary Digit.

**Binary explained**

1. Gottfried Leibniz (1646–1716) refined the binary number system in the 17th century. He invented it around 1679 and published it in 1701. Binary became the base of almost all modern computers. Leibniz also invented calculus.
2. A **bit** is the smallest unit of data in a computer. It has only one of two values: 1 or 0.
3. Inside the CPU (the computer’s “brain”) and RAM (its fast memory), these numbers appear as electrical signals: **on** (1) or **off** (0).

**Long Question: Describe how computers can save information using binary codes. [Exercise, Brief Q1]**

Computers save all information by storing it as patterns of 0s and 1s, using tiny electronic switches.

1. **Only two numbers:** A computer stores all information (text, pictures, sound, numbers) using only two digits: 0 and 1. This is binary code.
2. **Switches:** Millions of tiny electronic switches inside the computer are open (0) or closed (1).
3. **Bits and bytes:** Each switch is a bit (binary digit). Eight bits make one byte. Bits are stored in memory (RAM) and used by the CPU.
4. **Electrical signals:** In the CPU and RAM, these 1s and 0s are electrical signals: on or off.
5. **Codes for data:** Text is changed into binary using ASCII or Unicode (explained later). Pictures are stored as pixels with colours as RGB numbers. Sound is changed into numbers by an ADC.

**Conclusion:** More memory means more bits, and so more information, a computer can store.

**Importance of binary code**

The binary number system is the base of all computer systems. It lets a computer store, use and move information between the CPU and memory. Because of binary code, we can:

1. View websites.
2. Create and update documents.
3. Play games.
4. Watch streaming videos and other graphics (pictures).
5. Use software (apps and programs).
6. Do calculations and study data.

Using only 1s and 0s is an efficient way for a computer to work. It also helps control logic circuits, telling the computer whether a signal is true (1) or false (0).

---

## TOPIC 2: Binary and Decimal Number Systems

A **number system** is a way of writing numbers. Computers use the binary system; people use the decimal system in everyday life.

**Decimal number system (4.2)**

1. **Base:** It uses base 10, with the symbols 0 to 9.
2. **Other name:** It is also called the **Hindu-Arabic number system**.
3. **Place value:** Each place is ten times the one on its right. Example, the number **1,980**:

| Digit | 1 | 9 | 8 | 0 |
|---|---|---|---|---|
| Place | 1,000th place | 100th place | 10th place | 1st place |

So 1 = one thousand, 9 = nine hundred, 8 = eighty, 0 = zero ones.

| 1 | 9 | 8 | 0 |
|---|---|---|---|
| 1,000th place | 100th place | 10th place | 1st place |

1,980 = 1000 + 900 + 80 + 0

*Fig: place-value diagram for the decimal number 1,980*

4. **Decimal point:** It is used to write fractions (parts of a number).
5. **Writing:** It is written as $45_{10}$ or $118_{10}$. If no base is shown, the number is assumed to be decimal, the system people use most.

**Binary number system**

1. **Base:** It uses base 2, with only two digits: 0 and 1.
2. **Bit:** It is the smallest unit of data in a computer, either 1 or 0.
3. **Writing:** Binary numbers are written as $110_2$ or $10_2$.
4. **Use:** It is used in computer programming and coding, since a computer only understands 0 and 1.
5. **MSB and LSB:** The leftmost bit is the **Most Significant Bit (MSB)**; the rightmost bit is the **Least Significant Bit (LSB)**.

**Example:** Look at the binary number **1010**.

$1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 = 8 + 0 + 2 + 0 =$ **10**

So binary **1010** equals decimal **10**.

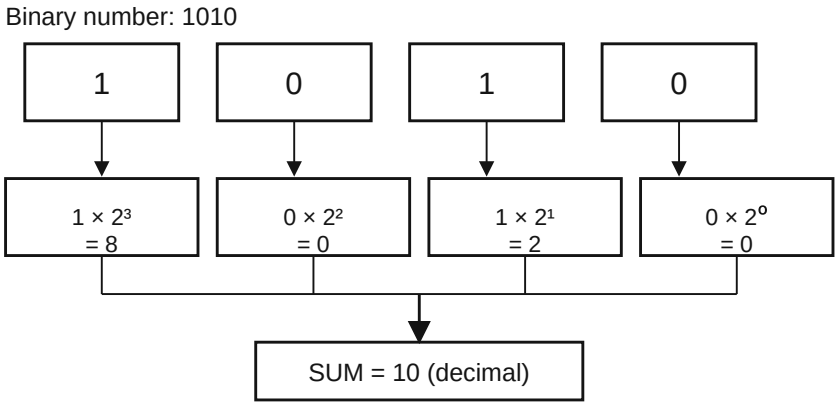



*Fig: binary place-value diagram showing $1010_2 = 10_{10}$*

### Units of memory (Table 4.1)

| Unit | Equal to |
|---|---|
| 1 Bit | Binary Digit (0 or 1) |
| 8 Bits | 1 Byte (B) |
| 1024 Bytes | 1 Kilobyte (KB) |
| 1024 KB | 1 Megabyte (MB) |
| 1024 MB | 1 Gigabyte (GB) |
| 1024 GB | 1 Terabyte (TB) |
| 1024 TB | 1 Petabyte (PB) |
| 1024 PB | 1 Exabyte (EB) |
| 1024 EB | 1 Zettabyte (ZB) |
| 1024 ZB | 1 Yottabyte (YB) |
| 1024 YB | 1 Brontobyte |
| 1024 Brontobyte | 1 Geopbyte |

### Memory units in decimal (about how big they are)

| Unit | Binary value | About this many bytes (decimal) |
|---|---|---|
| Bit | 0 or 1 | 1/8 of a byte |
| Byte | 8 bits | 1 byte |
| Kilobyte (KB) | $1024^1$ bytes | about 1,000 bytes |

| Unit | Binary value | About this many bytes (decimal) |
|---|---|---|
| Megabyte (MB) | $1024^2$ bytes | about 1,000,000 bytes |
| Gigabyte (GB) | $1024^3$ bytes | about 1,000,000,000 bytes |
| Terabyte (TB) | $1024^4$ bytes | about 1,000,000,000,000 bytes |
| Petabyte (PB) | $1024^5$ bytes | about 1,000,000,000,000,000 bytes |
| Exabyte (EB) | $1024^6$ bytes | about 1,000,000,000,000,000,000 bytes |
| Zettabyte (ZB) | $1024^7$ bytes | about 1,000,000,000,000,000,000,000 bytes |
| Yottabyte (YB) | $1024^8$ bytes | about 1,000,000,000,000,000,000,000,000 bytes |

**Remember:** In binary counting, 1 KB = $1024^1$ bytes, 1 MB = $1024^2$ bytes, and so on. In decimal counting, 1 KB is about 1,000 bytes, 1 MB about 1,000,000 bytes, and 1 GB about 1,000,000,000 bytes.

**Long Question: Differentiate between binary and decimal number systems. [Exercise, Brief Q2]**

Binary and decimal number systems differ mainly in their base and the digits they use.

| Binary Number System | Decimal Number System |
|---|---|
| It has base 2. | It has base 10. |
| It uses only two digits: 0 and 1. | It uses ten digits: 0 to 9. |
| Computers use it. | People use it in daily life. |
| Each place is 2 times the place on its right (1, 2, 4, 8 …). | Each place is 10 times the place on its right (1, 10, 100 …). |
| It is written with a base, like $1010_2$. | It is written like $10_{10}$, or with no base at all. |
| The leftmost bit is the MSB and the rightmost bit is the LSB. | It is also called the Hindu-Arabic number system. |

**Example:** The binary number $1010_2$ equals the decimal number $10_{10}$.

## TOPIC 3: Number System Conversion

We can change a binary number into a decimal number, and just as easily convert decimal into binary.

**Long Question: How does a computer convert binary numbers into decimal numbers? [Exercise, Brief Q3]**

We use a method called **positional notation**, where each digit's "weight" is the base (2) raised to a power based on its position.

1. **Step 1:** Write the power of 2 for each digit, starting from the rightmost place: $2^0$, then $2^1$, $2^2$, and so on.
2. **Step 2:** Multiply each digit by its power of 2.
3. **Step 3:** Add all the results. The sum is the decimal number.

**Example:** Convert $101101_2$ into decimal.

| Digit | 1 | 0 | 1 | 1 | 0 | 1 |
|---|---|---|---|---|---|---|
| Power of 2 | $2^5$ | $2^4$ | $2^3$ | $2^2$ | $2^1$ | $2^0$ |
| Value | 32 | 0 | 8 | 4 | 0 | 1 |

$1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 32 + 0 + 8 + 4 + 0 + 1 =$ **45**
So $\mathbf{(101101)_2 = (45)_{10}}$.

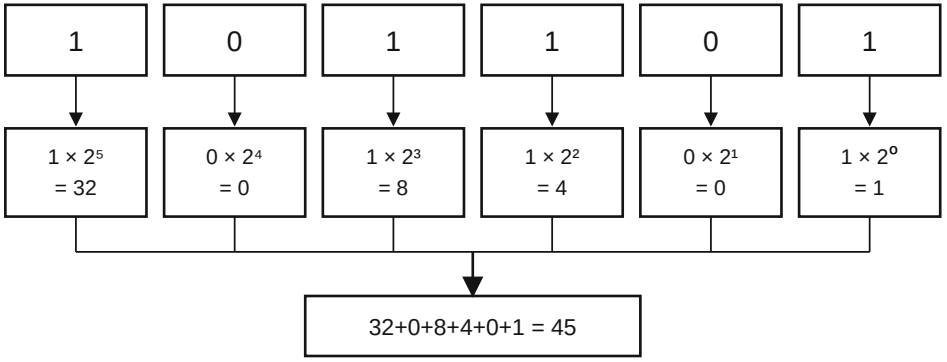



*Fig: binary-to-decimal worked diagram for $101101_2 = 45_{10}$*

**Long Question: How does a computer convert decimal numbers into binary numbers? [Exercise, Brief Q4]**

To change decimal into binary, divide the number by 2 repeatedly.

1. **Step 1:** Divide the decimal number by 2, and write down the remainder (0 or 1).
2. **Step 2:** Divide the quotient by 2 again, and write down the remainder.
3. **Step 3:** Keep repeating this until the quotient becomes 0.
4. **Step 4:** Write the remainders in reverse order, last one first. (The first remainder is the LSB; the last is the MSB.)

**Example 1:** Convert $100_{10}$ to binary.

| Divide by 2 | Number | Remainder |
|---|---|---|
| | 100 | |
| | 50 | 0 |
| | 25 | 0 |
| | 12 | 1 |
| | 6 | 0 |
| | 3 | 0 |
| | 1 | 1 |

| Divide by 2 | Number | Remainder |
|---|---|---|
| | 0 | 1 (stop here) |

Read the remainders from bottom to top: **1100100**. So **$(100)_{10} = (1100100)_2$**.

**Example 2:** Convert $14_{10}$ to binary.

14 ÷ 2 = 7 r 0 → 7 ÷ 2 = 3 r 1 → 3 ÷ 2 = 1 r 1 → 1 ÷ 2 = 0 r 1

Read bottom to top: **1110**. So **$(14)_{10} = (1110)_2$**.

**Example 3:** Convert $172_{10}$ to binary.

172 ÷ 2 = 86 r 0 → 86 ÷ 2 = 43 r 0 → 43 ÷ 2 = 21 r 1 → 21 ÷ 2 = 10 r 1 → 10 ÷ 2 = 5 r 0 → 5 ÷ 2 = 2 r 1 → 2 ÷ 2 = 1 r 0 → 1 ÷ 2 = 0 r 1

Read bottom to top: **10101100**. So **$(172)_{10} = (10101100)_2$**.

(Check: 128 + 32 + 8 + 4 = 172. Correct.)

**Decimal to binary chart (0 to 20)**

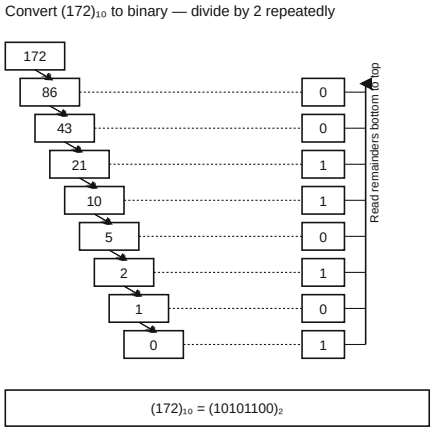



*Fig: decimal-to-binary staircase diagram for $172_{10}$ = $10101100_2$ (corrected — see Mistakes Found, Page 61)*

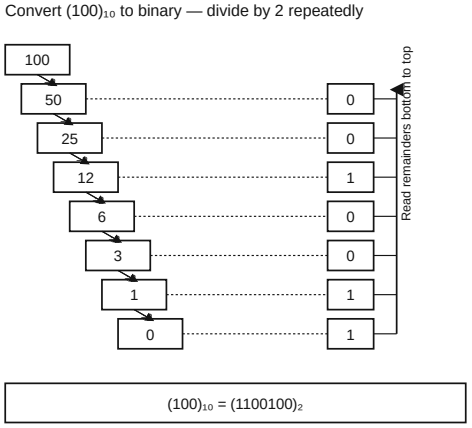



*Fig: decimal-to-binary staircase diagram for $100_{10} = 1100100_2$*

| Decimal | Binary | Decimal | Binary | Decimal | Binary |
|---|---|---|---|---|---|
| 0 | 0 | 7 | 111 | 14 | 1110 |
| 1 | 1 | 8 | 1000 | 15 | 1111 |
| 2 | 10 | 9 | 1001 | 16 | 10000 |
| 3 | 11 | 10 | 1010 | 17 | 10001 |
| 4 | 100 | 11 | 1011 | 18 | 10010 |
| 5 | 101 | 12 | 1100 | 19 | 10011 |
| 6 | 110 | 13 | 1101 | 20 | 10100 |

## TOPIC 4: Text, Image and Sound in Binary (4.3)

Text, pictures and sound must also be changed into binary before a computer can store and use them.

**Long Question: Write a note explaining how text, image and sound are represented in binary code. [Exercise, Brief Q5]**

Text, images and sound are each changed into binary numbers in their own way, so a computer can store and use them.

**1. Text**

1. Pressing a key sends a binary number to the computer, which displays the letter. Example: **A = 0100 0001**.
2. **ASCII** (American Standard Code for Information Interchange) is a code that gives a binary number to every character on the keyboard.
3. ASCII is a **7-bit code**, storing $2^7 = 128$ characters, enough for English but not other languages. It was later extended to $2^8 = 256$ characters, still not enough for every language.
4. **Unicode** was made to show more than 65,000 characters, so any language, like Russian or Chinese, can be used on a computer.

**Text examples (ASCII):**

1. Letter **a** = 0110 0001 (decimal 97)
2. Letter **b** = 0110 0010 (decimal 98)
3. Letter **c** = 0110 0011 (decimal 99)

ASCII codes start counting at 0 and also cover special characters, like punctuation marks, the Enter key, and control keys.

**2. Images**

1. A digital image is made of small dots called **pixels**. Each one is stored as a binary number.
2. In a black and white picture, 1 means black (on) and 0 means white (off).
3. Before drawing a picture, the computer must know the size of the pixel grid. This is called **metadata**. If the metadata says 10x10, the picture is 10 pixels across and 10 pixels down.

Example grid (1 = black, 0 = white):

| **0** | **0** | **0** | **0** | **0** | **0** | **0** | **0** |
|---|---|---|---|---|---|---|---|
| 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 1 | 0 | 0 | 0 | 0 | 1 | 0 |
| 0 | 0 | 1 | 1 | 1 | 1 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

Coloured pictures need more numbers. Colours are stored as 8-bit numbers, using a system called **RGB** (Red, Green, Blue).

**Colours in binary code**

1. **8-bit numbers:** Each colour part (Red, Green, Blue) is stored using 8 bits, giving 256 levels (0 to 255). Example: 0 = 00000000, 255 = 11111111.
2. **RGB:** Colours on a screen are made by mixing Red, Green and Blue light, each with a value from 0 to 255.

3. **Primary colours:** They are Red (255,0,0), Green (0,255,0), Blue (0,0,255).
4. **Secondary colours:** They are Cyan (0,255,255), Magenta (255,0,255), Yellow (255,255,0).
5. **Black and white:** Black is (0,0,0), white is (255,255,255), and medium gray is (128,128,128).
6. **Hexadecimal:** Some programs show colours using hexadecimal, a 16-symbol system: 0 to 9, then A to F.

**Colours in decimal, binary and hexadecimal**

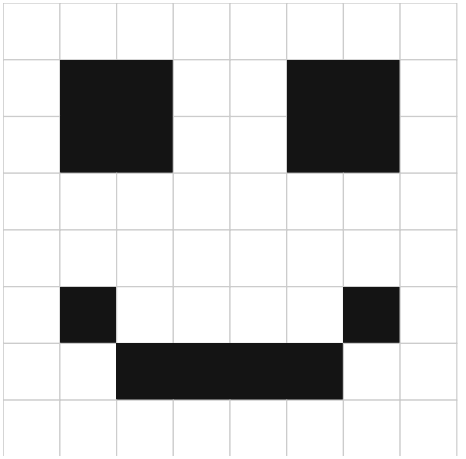


*Fig: the black-and-white pixel-grid picture drawn from the 8x8 grid above*

| Colour | Decimal RGB | Binary RGB | Hexadecimal RGB |
|---|---|---|---|
| Black | (0,0,0) | (00000000, 00000000, 00000000) | (00,00,00) |
| White | (255,255,255) | (11111111, 11111111, 11111111) | (FF,FF,FF) |
| Red | (255,0,0) | (11111111, 00000000, 00000000) | (FF,00,00) |
| Green | (0,255,0) | (00000000, 11111111, 00000000) | (00,FF,00) |
| Blue | (0,0,255) | (00000000, 00000000, 11111111) | (00,00,FF) |
| Cyan | (0,255,255) | (00000000, 11111111, 11111111) | (00,FF,FF) |
| Magenta | (255,0,255) | (11111111, 00000000, 11111111) | (FF,00,FF) |
| Yellow | (255,255,0) | (11111111, 11111111, 00000000) | (FF,FF,00) |
| Gray | (128,128,128) | (10000000, 10000000, 10000000) | (80,80,80) |
| Pale Yellow | (200,180,120) | (11001000, 10110100, 01111000) | (C8,B4,78) |

The screen decides what colour we see. The same numbers, like (200,180,120), may look slightly different on different screens or printers, making exact colours a hard technical problem.

**3. Sound**

1. Sound waves are **analogue**: smooth, continuous waves, not numbers.
2. To store sound on a computer, the wave must be changed into numbers (digital data) by an **ADC** (Analogue-to-Digital Converter).
3. The ADC works by taking “samples” (readings) of the sound wave at regular, equal time gaps.
4. **Sample rate:** how many samples are taken every second. More samples mean better sound quality.
5. **Bit rate:** how many bits are used to store each sample. More bits mean better accuracy, but a bigger file size.

6. **Audacity** is a free audio editor program. We can open an MP3 file in it, zoom in to see each sample's value, and save the sound at different sample rates to hear the quality change.

## TOPIC 5: Scratch Programming

**Scratch** is a programming software that makes it easy to build interactive games, animations and stories.

### Scratch and its interface (4.4)

When Scratch opens, it shows one character by default, called a **sprite** (the Scratch Cat). We write code to make a sprite move and interact with the user.

### Parts of the Scratch interface

| Part | What it does |
|---|---|
| Block Categories | Shows groups of blocks (Motion, Looks, Sound, Events, Control, Sensing, Operators, Variables). Click a category to see its blocks. |
| Block Palette | Shows the list of blocks. Click a block and drag it to the script area to use it. |
| Script Area | The place where we join blocks together to build a program. |
| Stage | The place where the program runs. We see the sprite move here. |
| Sprite List | Shows all the sprites in the program. We can add, delete or change sprites and the backdrop (background) here. |
| X and Y Coordinates | Show the exact position of the sprite on the stage. (0, 0) is the middle of the stage. |

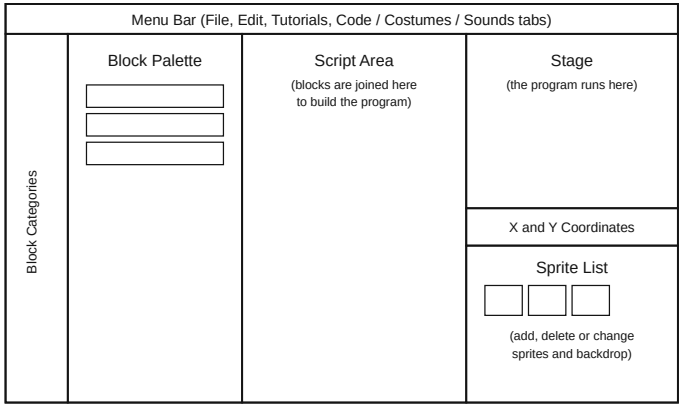



*Fig: layout of the Scratch programming interface*

**Other parts of the screen:** Code tab (shows the Block Palette), Costumes tab (a paint editor), Sounds tab (a sound editor), Menu bar, Stage Size buttons (to make the stage bigger or smaller), Choose a Sprite button, Choose a Backdrop button, and Extensions Library.

### Combination of programming constructs (4.4.1)

We can combine **events** and **coordinates** (position numbers) so a sprite can move in two ways: 1. The user moves it by pressing keys. 2. The sprite moves by itself (automatically), with no help from the user.

1. **Movement:** The most basic thing a game does, usually the first thing added to a new game. In a game, players press arrow keys to move their character, letting it interact with the game world.
2. **Step 1 — Select a sprite:** Every object in a Scratch project is a sprite, a character that can move and run code. The Scratch Cat is in every new project by default. To add a new one, click **Choose a Sprite** at the bottom right, which opens the Sprite Menu, a library of ready-made characters; clicking one adds it to the project.
3. **Step 2 — Program the sprite:** The easiest way to make a sprite move is with **event listeners**, blocks that wait for something to happen, like a key press. The code pairs a yellow event block (tells the sprite *when* to move) with blue motion blocks (move the sprite).

**Tip:** Right-click a block and choose **Duplicate** to copy it, instead of dragging new blocks each time. (The right-click menu also has Add Comment and Delete Block.)

### Moving in four directions

*"When you press the right arrow key, point towards the right, then move forward 10 steps."*

1. Each script runs when a different arrow key is pressed.
2. The **point in direction** block uses degrees to set direction: 90 = right, -90 = left, 0 = up, 180 = down.
3. The **move 10 steps** block is the same in all four scripts, so the sprite always moves at the same speed.

**Step method (the sprite turns, then moves):**

| Key pressed | point in direction | move |
|---|---|---|
| right arrow | 90 | move 10 steps |
| left arrow | -90 | move 10 steps |
| up arrow | 0 | move 10 steps |
| down arrow | 180 | move 10 steps |

**X-Y method (the sprite moves, but does not turn):**

| Key pressed | Block used |
|---|---|
| right arrow | change x by 10 |
| left arrow | change x by -10 |
| up arrow | change y by 10 |
| down arrow | change y by -10 |

### Long Question: Allow the user to move a sprite using arrow keys. [Exercise, Project Q4]

There are three ways to make a sprite move right, left, up and down using the arrow keys:

1. **X-Y Method:** Select the sprite. Use “when [arrow] key pressed” with:
   - “change x by 10” for the right arrow key
   - “change x by -10” for the left arrow key
   - “change y by 10” for the up arrow key
   - “change y by -10” for the down arrow key

   The sprite moves but always keeps pointing in the same direction.
2. **Step Method:** Use “when [arrow] key pressed”, then “point in direction” (90 for right, -90 for left, 0 for up, 180 for down), then “move 10 steps”. The sprite turns to face the new direction, then moves. Not ideal if the sprite needs to turn for other reasons too.
3. **Loop Method:** Put either method inside a **forever** loop with “if key pressed” blocks, so the program keeps checking the keys continuously, not just once.

---

## TOPIC 6: Variables and Loops in Scratch

**Variables** help a program remember a value, while **loops** repeat the same blocks again and again.

### Variables

Sometimes a program must remember a value to use many times. The best way is a **variable**, a named place that holds a value.

1. A variable is given a name, used to call it in the program.
2. A variable is assigned a value: a word (string), a number, or a letter.
3. A variable’s value can be changed as many times as needed.
4. As many variables as needed can be made.

### To create a variable in Scratch:

1. Go to the Variables category.
2. Click **Make a Variable**.
3. A “New Variable” box appears. Type a name for it.
4. Choose “For all sprites” or “For this sprite only”.
5. Click OK.

Now blocks like **set**, **change**, **show variable** and **hide variable** appear, ready to use.

1. **Set block:** “set [variable] to (0)” gives the variable a starting value. Click the small arrow to pick the variable; it can be set to a number (like 12) or a word (like “Hello”). Always give a variable a value before using it.
2. **Reporter block:** The round (variable) block shows the value stored in the variable. Example: “say (message1) for 2 seconds” shows the value stored in message1 on the stage.

### Using variables with random numbers

The **pick random** block gives a random (unpredictable) number within a chosen range. In the script below, Number1 and Number2 each get a random number from 1 to 11, and the sprite says their product (the answer when multiplied).

```
when green flag clicked
set Number1 to (pick random 1 to 11)
```

```
set Number2 to (pick random 1 to 11)
say (join "The product of both numbers is: " (Number1 × Number2)) for 2 seconds
```

To show the two random numbers alongside the product, join more text blocks together (nested join blocks):

```
when green flag clicked
set Number1 to (pick random 1 to 11)
set Number2 to (pick random 1 to 11)
say (join "The product " (join Number1 (join " and " (join Number2 (join " is: " (Number1 × Number2)))))) for
2 seconds
```

**Working with loops in Scratch**

The Control category has two kinds of loop blocks:

1. **Repeat Loop:** It repeats the blocks inside it a fixed number of times, set at the start of the loop.
2. **Forever Loop:** It keeps repeating the blocks inside it until something stops it (like a “stop all” block inside an “if” condition). Without a stopping condition, it becomes an **infinite loop** that never stops on its own.

**Program Example: Turn a sprite by 15 degrees, 20 times, with a 0.25 second delay each time.**

```
when green flag clicked
repeat 20
   wait 0.25 seconds
   turn right 15 degrees
```

**Program Example (Learning Outcome): Create a countdown timer in Scratch.**

Make a timer that starts at 100 and stops the program at 0. The value should change every 0.5 seconds.

1. Make a variable named **Timer**.
2. Use “when green flag clicked” to start the program.
3. Use “set Timer to 100” to give it a starting value.
4. Add a **forever** loop.
5. Inside it, use “if Timer < 1 then” → “stop all” to stop the program once the timer reaches 0.
6. In the “else” part, use “wait 0.5 seconds” and “change Timer by -1” to decrease the timer by 1 every half second.

```
when green flag clicked
set Timer to 100
forever
   if Timer < 1 then
      stop all
   else
      wait 0.5 seconds
      change Timer by -1
```

**Long Question: Create a game with a player sprite and a food sprite, using coordinates, conditionals, loops and variables. [Exercise, Project Q3, Q5, Q6]**

This game uses a player sprite that chases a food sprite, scoring a point each time they touch.

1. **Two sprites:** Add a player sprite and a “food” sprite. The player’s job is to chase the food.
2. **Move the player (events + coordinates):** Use “when [arrow] key pressed” with “change x by” or “change y by” blocks, so the player moves right, left, up and down with the arrow keys.
3. **Move the food automatically:** In the food sprite, use “when green flag clicked”, then a “forever” loop, then “glide 1 secs to random position” (or “go to random position”), so the food moves by itself, with no help from the user.
4. **Score variable:** Make a variable called **Score**, and set it to 0 when the game starts.
5. **Conditional (if-then check):** Inside a forever loop, use “if touching Food? then change Score by 1”, so the player earns a point each time it touches the food.

```
Player sprite (score):
when green flag clicked
set Score to 0
forever
   if touching Food? then
      change Score by 1
```

```
Food sprite (moves by itself):
when green flag clicked
forever
   glide 1 secs to random position
```

## TOPIC 7: Programming Activities (Exercise Project Questions 1 and 2)

**Long Question: Give instructions to complete a simple task, like drawing a cat or a dog. [Exercise, Project Q1 / Activity Q1]**

This activity teaches us that a computer needs exact, step-by-step instructions. Humans can guess what we mean, but a computer cannot.

1. Work in pairs. Each student gets one piece of paper and a pencil.
2. Student 1 draws a simple animal, like a cat, without showing it to Student 2.
3. Student 1 tells Student 2 how to draw the same picture, using only words (instructions), not hand signs or pointing.
4. Student 2 must follow the instructions exactly, step by step, to try to draw the same picture.
5. At the end, compare both drawings.

**What we learn:** A computer can only follow clear, exact instructions written in a programming language. If an instruction isn’t clear enough, the computer (or Student 2) can’t do the task correctly. This is why programming needs standard, exact terms.

**Long Question: Break down a computer, phone or online game into simple instructions in your own language. [Exercise, Project Q2 / Activity Q2]**

Pick any game you’ve played, and answer these questions to break it into simple instructions:

1. What is the objective (goal) of the game?
2. How many sprites (characters or objects) are there in the game?
3. How does the player move?
4. How does the player win (if there is a way to win)?
5. How does the player lose (if there is a way to lose)?

6. What instructions have been given to the computer to make this game work?

**Example (Fruit Catching game):**

1. **Objective:** It is to catch as many falling fruits as possible before they hit the ground.
2. **Sprites:** There are two sprites: a basket (the player) and a fruit.
3. **Movement:** The player moves the basket left and right using the arrow keys.
4. **Win:** The player "wins" by reaching a high score, such as catching 20 fruits.
5. **Lose:** The player loses points, or the game ends, if a fruit falls past the basket and hits the ground.
6. **Instructions given to the computer:** Move the basket when a key is pressed; make the fruit fall automatically from the top; check if the basket touches the fruit; add 1 to the score if it touches; end the game if the fruit reaches the bottom without being caught.

## SHORT QUESTIONS — UNIT 4

1. **Describe how computers save information using binary codes.** *[Topic 1 · Exercise]*
   Computers store all information as 0s and 1s. Tiny switches inside the computer are off (0) or on (1). Each switch is a bit, and 8 bits make a byte.
2. **Differentiate between binary and decimal number systems.** *[Topic 2 · Exercise]*
   The binary system has base 2 and uses 0 and 1. The decimal system has base 10 and uses 0 to 9. Computers use binary. People use decimal.
3. **How does a computer convert binary numbers into decimals?** *[Topic 3 · Exercise]*
   Multiply each binary digit by its power of 2 ($2^0$, $2^1$, $2^2$ ... counting from the right). Then add all the results. Example: $101101_2 = 45$.
4. **How does a computer convert decimal numbers into binary?** *[Topic 3 · Exercise]*
   Divide the number by 2 again and again. Write down each remainder, until the quotient becomes 0. Then write the remainders from the last one to the first one. Example: $14_{10} = 1110_2$.
5. **Write a note on text representation in binary.** *[Topic 4 · Exercise]*
   Each key is changed into a binary code using ASCII or Unicode. Example: A = 0100 0001. ASCII can store 128 characters. Unicode can store more than 65,000 characters.
6. **What is a bit?** *[Topic 1 · Book concept]*
   A bit (binary digit) is the smallest unit of data in a computer. It is either 0 or 1.
7. **What is a byte?** *[Topic 1 · Book concept]*
   Eight bits together make one byte.
8. **Who refined the binary number system?** *[Topic 1 · Book concept]*
   Gottfried Leibniz refined the binary system in the 17th century. He invented it around 1679 and published it in 1701.
9. **What are MSB and LSB?** *[Topic 2 · Book concept]*
   In a binary number, the leftmost bit is the Most Significant Bit (MSB). The rightmost bit is the Least Significant Bit (LSB).

## MULTIPLE CHOICE QUESTIONS — UNIT 4

These are the book's own Exercise MCQs plus a few extra practice MCQs, with the correct answer marked ✔ above the option.

1. The term __ also refers to any digital encoding/decoding system in which there are exactly two possible states. *[Topic 1 · Exercise MCQ 1]*

| a) Digital | **b) Binary** ✔ | c) Python | d) Programming |
|---|---|---|---|

2. __ bits together make what we call a "byte" of data. *[Topic 1 · Exercise MCQ 2]*

| a) 6 | b) 7 | **c) 8** ✔ | d) 9 |
|---|---|---|---|

3. To convert numbers from decimal to binary, the given decimal number is divided repeatedly by __. *[Topic 3 · Exercise MCQ 3]*

| | | | |
|---|---|---|---|
| ✔ **a) 2** | b) 3 | c) 4 | d) 5 |

4. __ are made up of pixels. *[Topic 4 · Exercise MCQ 4]*

| | | | |
|---|---|---|---|
| a) Binary numbers | b) Decimals | c) Programs | ✔ **d) Digital images** |

5. ASCII code can only store up to __ characters. *[Topic 4 · Exercise MCQ 5]*

| | | | |
|---|---|---|---|
| ✔ **a) 128** | b) 256 | c) 64 | d) 124 |

6. __ are represented in a computer using 8-bit numbers. *[Topic 4 · Exercise MCQ 6]*

| | | | |
|---|---|---|---|
| a) Numbers | b) Letters | c) Images | ✔ **d) Colours** |

7. We need __ to convert sound waves into numerical representation. *[Topic 4 · Exercise MCQ 7]*

| | | | |
|---|---|---|---|
| a) Audio to digital converter | b) Audio to digits converter | ✔ **c) Analogue to digital converter** | d) Analogue to digits converter |

8. The best way to remember a value when designing a program is to use a __. *[Topic 6 · Exercise MCQ 8]*

| | | | |
|---|---|---|---|
| ✔ **a) Variable** | b) Value | c) Number | d) Binary |

9. The __ is/are analogue. *[Topic 4 · Exercise MCQ 9]*

| | | | |
|---|---|---|---|
| a) Light | b) Numbers | c) Values | ✔ **d) Sound** |

10. The decimal number system represents a number with a base of __. *[Topic 2 · Exercise MCQ 10]*

| | | | |
|---|---|---|---|
| a) 2 | b) 3 | c) 5 | ✔ **d) 10** |

11. Bit is the short form of: *[Topic 1]*

| | | | |
|---|---|---|---|
| a) Binary Item | ✔ **b) Binary Digit** | c) Basic Digit | d) Byte Digit |

12. The smallest unit of data in a computer is: *[Topic 1]*

| | | | |
|---|---|---|---|
| a) byte | ✔ **b) bit** | c) nibble | d) kilobyte |

13. The binary number system was refined by: *[Topic 1]*

| | | | |
|---|---|---|---|
| a) Charles Babbage | ✔ **b) Gottfried Leibniz** | c) Bill Gates | d) Isaac Newton |

14. In a computer, a closed (on) switch represents: *[Topic 1]*

| | | | |
|---|---|---|---|
| a) 0 | ✔ **b) 1** | c) 2 | d) 10 |

## Mistakes Found in the Book

1. **Page 61:** The picture shows $(172)_{10} = (1010100)_2$, but this is wrong: $1010100_2$ equals only 84. Using the book's own division steps on the same page, the correct answer is $(172)_{10} = (10101100)_2$.
2. **Page 72:** The text says the script shows "the product" of Number1 and Number2, but the picture places them side by side, with no multiply (×) sign between them. The script should multiply the two numbers to get the real product.
3. **Page 73:** In the four arrow-key scripts, several blocks say "up arrow" when they should say other directions (left, down), and the step-method scripts also say "move 10 stage" instead of "move 10 steps". The correct scripts, with the right key names and block name, are shown in these notes (Topic 5).
4. **Page 74:** Under the heading "Repeat Loop", the book shows a picture of the empty **forever** block, not the **repeat** block. The forever block picture belongs under "Forever Loop" instead.

5. **Page 57:** The book says a gigabyte is “approximately trillions of bytes” — this is wrong. A gigabyte is about one **billion** bytes, not a trillion.
6. **Page 58, Table 4.1:** The table spells “Terabyte” as “Terra Byte”. The correct spelling is “Terabyte”.
7. **Page 59:** The memory table’s column heading is spelled “Abrivation”. The correct spelling is “Abbreviation”.
8. **Pages 65–66:** The Sprite and Scratch Interface figures are captioned “Fig. 5.4 Sprite” and “Fig. 5.3 Scratch Interface”. Since this is Unit 4, the figure numbers should start with 4 (like “Fig. 4.x”), not 5.
9. **Page 77, Exercise MCQ 3:** The question says a decimal number is divided by 2 “till we get 0 or 1 as the final quotient”. But the book’s own worked method (page 61) stops only when the quotient becomes 0, not 0 or 1. The correct rule is: keep dividing until the quotient is 0.

# Unit 5 — Digital Citizenship

## TOPIC 1: ETHICS AND ETHICAL ISSUES

Ethics tells us what is right and wrong, and guides how we behave with others.

### Ethics (5.1)

**Ethics** is the study of what is morally good and bad, or right and wrong. The word “ethics” is also used for a system of moral values or rules.

Ethics is based on clear standards, which tell people: 1. Their rights. 2. Their duties (things they must do). 3. How their actions help society. 4. How to be fair. 5. How to have good qualities, like honesty (these qualities are called “virtues”).

**Extra Bit:** A Greek thinker named Aristotle said ethics is the study of human relations in their most perfect form. He called it “the science of proper behaviour.”

Six values build ethics for success: Honesty, Integrity, Responsibility, Quality, Trust and Respect, and Leadership.

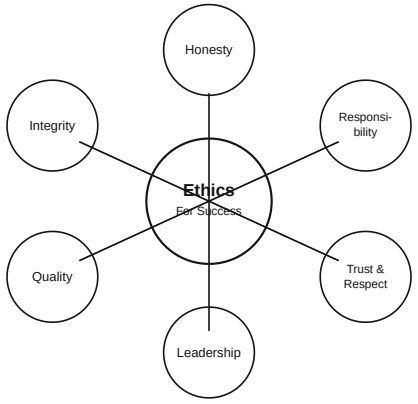



*Fig: Ethics for success — six values arranged around “Ethics”*

### Ethical issues

An **ethical issue** is a situation where a moral problem comes up in society or at work, challenging or breaking a moral rule.

Ethical issues include: 1. Privacy and confidentiality (keeping information secret) 2. Issues related to socially vulnerable (weak) people 3. Health insurance discrimination (treating some people unfairly about health insurance) 4. Employment (job) discrimination 5. Individual responsibility 6. Issues related to race and ethnicity 7. Implementation of laws (making sure laws are followed)

## TOPIC 2: ETHICS IN THE DIGITAL WORLD (5.2)

Today, most of our relationships have moved online: we chat, message, and use social media to talk to people.

### What is digital ethics?

We live in a digital world, where most of our relationships have moved online, to chats, messengers, and social media. Knowing and following the rules of good online behaviour is called **digital ethics**.

### Ethical issues in the digital world

Technology has given us many benefits: 5G, Virtual Reality, Mixed Reality, Artificial Intelligence, driverless vehicles, digital assistants, robots, smart devices, and automation.

But technology also has bad effects. Some common ethical issues in the digital world are: - Misuse of personal information - Misinformation (wrong information) and deep fakes (fake videos or pictures made to look real) - Lack of oversight (not enough checking) and not accepting responsibility - Social and political instability

Good online behaviour means respecting others, protecting privacy, not copying others' work, and not spreading false news.

### Long Question: Define ethics in the digital environment. *(Exercise – Detail Q1)*

Digital ethics means knowing and following the rules of good online behaviour, especially as technology brings new benefits and new risks.

1. **Meaning:** We live in a digital world, where most of our relationships have moved online, to chats, messengers, and social media. Knowing and following the rules of good online behaviour is called digital ethics.
2. **Benefits of technology:** Technology gives us many benefits, like 5G, Virtual Reality, Mixed Reality, Artificial Intelligence, driverless vehicles, digital assistants, robots, smart devices, and automation.
3. **Negative impacts:** Technology also has bad effects. Common ethical issues in the digital world are:
    - Misuse of personal information
    - Misinformation and deep fakes
    - Lack of oversight and not accepting responsibility
    - Social and political instability
4. **Good behaviour:** We should respect others online, protect privacy, avoid copying others' work, and never spread false news.

### Being a safe and responsible digital citizen

We spend most of our time online, and store most of our things there too: photos, documents, and more. So it is important to protect our devices and information from harmful people.

If we leave our devices unprotected, problems can follow: a slower computer, loss of money from a bank account, or identity theft (someone stealing your identity and pretending to be you).

A good **digital citizen** (a person who uses the internet safely and responsibly) does the following: 1. Encourages positive and healthy talk online. 2. Knows about common online scams and bad (toxic) behaviour. 3. Avoids cyberbullying. 4. Shows empathy (care and kindness) for others online.

### Long Question: Discuss the importance of being a safe and responsible digital citizen. *(Exercise – Detail Q2)*

Being a safe digital citizen matters because we spend most of our time online and keep most of our things there, so protecting our devices and information protects us too.

1. **Why it is important:** We spend most of our time online, and store most of our things there too. So it is important to protect our devices and information from harmful people.
2. **Danger of being unsafe:** If we leave devices unprotected, the result can be small, like a slower computer, or big, like losing all the money in a bank account, or identity theft.

3. **A good digital citizen:** encourages positive and healthy talk online, knows about common online scams and toxic behaviour, avoids cyberbullying, and shows empathy for others online.
4. **Characteristics of all good digital citizens:** They protect private information for themselves and others, respect themselves and others, and stay safe online. They also stand up to cyberbullying, balance the time they spend online, respect copyright and intellectual property (other people's creative work, like songs or books), and manage their digital footprint (the trail of information they leave behind online) carefully.

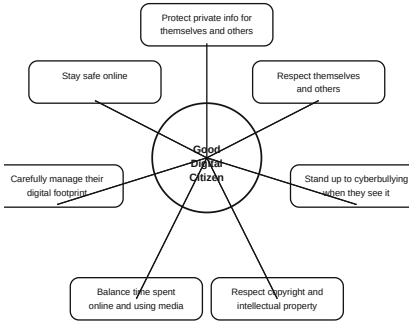



*Fig: Characteristics of a good digital citizen*

**Responsible digital citizenship**

**Long Question: What do you mean by a responsible digital citizen?** *(Exercise – Detail Q8)*

A responsible digital citizen treats people and information online with the same respect and care they would offline.

1. **Responsible digital citizenship:** We must understand what happens when we post photos, videos, or other personal content. Once something is online, it is hard to remove, and it can become part of our permanent online reputation (how people see us). Photos can also be changed or shared without our permission.
2. **Respectful digital citizen:** Respect for ourselves and others matters in every relationship, and it is the same online. We should treat online friends with as much respect as face-to-face friends, and never create or forward nasty or humiliating emails, pictures, or messages about anyone.
3. **Online social skills:** A responsible digital citizen has the online social skills to take part in online community life in an ethical and respectful way.
4. **Properties of a good digital citizen:**
    1. Respect the rights of others.
    2. Practise tolerance (accept people who are different from you).
    3. Be informed about the world around you.
    4. Respect the property of others.
    5. Be compassionate (kind).
    6. Take responsibility for your actions.
    7. Pay taxes.

**Extra Bit:** If you get nasty or bullying comments, block or unfriend the people who do not treat you with respect.

### Protecting your online reputation

**Long Question: How can we protect our reputation online?** *(Exercise – Detail Q3)*

Being a responsible digital citizen means having the online social skills to take part in online community life in an ethical and respectful way. To protect our reputation, we should:

1. **Behave lawfully:** Follow the law when we are online.
2. **Protect privacy:** Protect our own privacy and the privacy of others.
3. **Know our rights and responsibilities:** Learn our rights and responsibilities when we use digital media.
4. **Think before acting:** Think about how our online activities affect ourselves, people we know, and the wider online community.
5. **Post carefully:** Think before posting photos and videos. Once they are online, they are hard to remove. They become part of our permanent online reputation.

### Protect your privacy

Ways to protect your privacy online: 1. Share only as much personal information as necessary. 2. Keep privacy settings up to date on social media sites. This way, your profile is not open to everyone. 3. Keep your passwords private. 4. Check the location settings and services on smartphones, tablets, and apps.

### Watch out for scams

Cybercriminals (criminals who use computers and the internet) can trick, or **scam**, people online in many ways. Most scams can be avoided if you have the right knowledge and tools. Three common scams:

1. **Phishing:** A scammer sends a malicious (harmful) link. The link is made to trick a person into giving away sensitive (private) information, like a password. The attacker can then watch everything the victim does on the site, and even get past extra security steps.
2. **Fake websites:** Cybercriminals make a fake site, usually of a shopping website, to trick you into entering your personal information. Then they misuse it.
3. **Tech support scams:** Fake “tech support” agents ask for your login information or remote access to your computer. Then they ask you for money to “fix” it.

**Long Question: How can we say that tech support scams are harmful?** *(Exercise – Detail Q7)*

Tech support scams are harmful because they trick victims step by step into handing over information, access, and money.

1. **Fake agents:** In a tech support scam, a person pretends to be a computer expert. Often a fake warning appears on the screen, such as: “System may have detected viruses on your computer. Call tech support right away.”
2. **Stealing information:** The fake agent asks for our login information (username and password), and can then open our accounts.
3. **Remote access:** He asks for remote access to our computer, so he can see our files, steal our data, or install harmful software.
4. **Asking for money:** After that, he asks for money to “fix” the computer. This is a red flag (warning sign). A real tech support agent would not usually ask for payment like this.

**Conclusion:** Tech support scams are harmful because we can lose our private data, our money, and the safety of our computer. We should never give passwords or remote access to strangers.

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## TOPIC 3: USES OF THE INTERNET

Using the internet well makes life easier, giving us useful data, information, and knowledge that helps us grow personally, socially, and economically (in terms of money and work).

### Use of internet in business (5.3.1)

The internet has become an important part of business, used for much more than email and a basic website. Common uses are:

1. **E-commerce (5.3.2):** It means buying and selling things online. According to the U.S. Census Bureau, it made up 14 percent of all retail sales in 2020. Government services are now online too, and businesses like Amazon and Shopify sell and deal with customers entirely on the internet.
2. **Marketing and CRM (5.3.3):** It means using **CRM (Client Relationship Management)** software, which uses data to help businesses serve customers better, sell more, and find new customers. CRM analytics lets a business treat each customer personally, instead of treating everyone the same. Tools like Facebook Pixel can find new customers by looking at the websites they visit, the posts they like, and the videos they watch on YouTube.
3. **Collaboration and document storage (5.3.4):** It means using tools like Google Drive, Microsoft OneDrive, and Dropbox to work together on projects in real time, from anywhere: updating, editing, and commenting on documents, while the file's creator controls who can see and edit it.
4. **Videoconferencing, chat, and remote employees (5.3.5):** It means businesses letting people work from home (remote work) to save the cost of a big office, using video calls, chat apps, and social media to keep teams connected, while automated chatbots answer customer questions quickly without a person waiting online.

**CRM: from leads to customers.** A lead moves through two stages: 1. Raw Leads → Viable Leads → Nurtured Leads → Active Leads, handled by **Marketing Automation**. 2. Marketing Qualified Leads → Sales Accepted Leads → Opportunities → Closed/Won, handled by **CRM**.

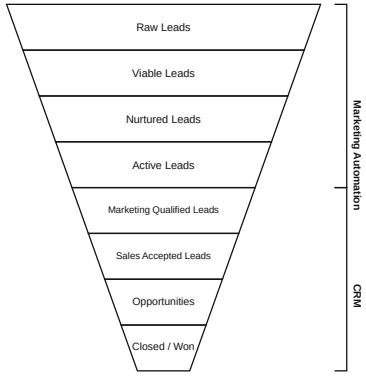



*Fig: CRM — from raw leads to closed customers*

### Use of internet in social networking (5.3.6)

**Social networking** means using internet-based social media sites to stay connected with friends, family, colleagues, or customers. It can serve a social purpose, a business purpose, or both. Examples are Facebook, Twitter, Instagram, and Pinterest. Social networking means building and keeping personal and business relationships using technology.

### Use of internet in entertainment (5.3.7)

We can enjoy a lot of content on the internet, through our phone, TV, or computer. Popular free online entertainment includes: 1. YouTube videos. 2. Podcasts (audio shows you can listen to online). 3. Music streaming, such as Pandora or Spotify. 4. News sites, such as ARY, Geo, Bol News, Dunia News, and National Public Radio. 5. Game sites, such as King.com and Miniclip.

## TOPIC 4: COPYRIGHT, PLAGIARISM AND PIRACY

We must respect other people's work. Copying it without permission is wrong, and can even be a crime.

### Copyright (5.4)

**Copyright** is a type of **intellectual property right** (intellectual property is something a person creates with their mind, like a song, a book, or an invention). Copyright protects the original work of its owner.

Under copyright law, protected works include: paintings, photographs, illustrations, musical compositions, sound recordings, computer programs, books, poems, blog posts, movies, architectural works, and plays.

### Plagiarism (5.5)

**Plagiarism** means presenting someone else's work or ideas as your own. It is still plagiarism even with their permission. This covers all published and unpublished material, electronic or printed. Plagiarism counts as academic dishonesty (cheating in schoolwork) and is a disciplinary offence (a punishable act).

**Types of plagiarism:** 1. **Global plagiarism:** Copying a whole text. 2. **Paraphrasing plagiarism:** Rewriting someone else's ideas in new words and presenting them as your own thoughts. 3. **Patchwork plagiarism:** Copying phrases and ideas from different sources and joining them into a new text. 4. **Self-plagiarism:** Using your own old work again, when you have already submitted or published it before.

### Piracy (5.6)

**Piracy** means illegally copying copyrighted material, such as books, computer programs, and films. It is a criminal act, aimed at stealing other people's ideas and inventions.

**Difference between piracy and plagiarism:**

| Plagiarism | Piracy |
|---|---|
| Showing someone else's work or ideas as your own | Illegally copying, using, or selling copyrighted material |
| The person takes credit for another person's work | The person copies the work without permission or payment |
| It is academic dishonesty and a disciplinary offence | It is a crime (a criminal act) |
| Example: copying an essay from the internet and putting your own name on it | Example: copying a movie or software and selling it |

## TOPIC 5: DIGITAL MEDIA BIAS

Media bias happens when a news channel lets its own opinions affect the news it reports.

### Digital media bias (5.7)

**Media criticism** means examining and studying the messages in mass media (TV, newspapers, websites, and so on). A media channel may show **bias** (unfairness, favouring one side) in how it reports a news story, or in which stories it chooses to cover as more important.

**Types of media bias:** 1. Slant Bias 2. Spin Bias 3. Sensationalism Bias (making news sound more exciting or scary than it is) 4. Story Choice Bias (choosing to cover some stories and not others) 5. Word Choice Bias 6. Omission Bias (leaving out important information)

**Long Question: Discuss the impacts of media bias.** *(Exercise – Detail Q6)*

Media bias affects what news we see and how we understand it, in several harmful ways.

1. **One-sided news:** Media bias shows up in which news stories get chosen, and how they're covered, so people end up getting only one side of the story.
2. **Wrong opinions:** When a news channel lets its opinions affect the news, people may form wrong opinions about people, groups, or events.
3. **Hidden information:** Because of omission bias and story choice bias, some important news is not shown at all.
4. **Exaggeration:** Sensationalism bias makes small news look much bigger and more exciting than it is, which can create fear or anger in people.
5. **Hiding criticism:** Calling a media source "unbiased" can also be a trick to hide bias, or to stop criticism, debate, and discussion.

### Importance of unbiased media (5.8)

Unbiased news is valuable. News channels should give a balanced view, collecting views from all sides of politics, understanding different worldviews, and avoiding media bias.

Unbiased news fairly shows many points of view on topics where people disagree. Biased news from one side (such as a conservative view) is still part of the media, and that's fine, as long as the channel is open and honest about its bias.

**Long Question: How can we avoid media bias?** *(Exercise – Detail Q4)*

To avoid the problems of media bias: 1. Give the reader multiple views on news and issues. 2. Strengthen our media literacy skills (our ability to understand and judge media). 3. Try to get all sides of an issue. 4. Watch different news channels that have different views. 5. Remember: if it sounds too good to be true, it probably is not true. 6. Read the footnotes and small warnings (caveats) in advertisements or promotions.

## TOPIC 6: MISUSE OF COMPUTER RESOURCES AND ONLINE PRIVACY

Computer resources must be used the right way. We must also keep our personal information private and safe.

### Misuse of computer resources (5.9)

**Long Question: List the misuse of computer resources.** *(Exercise – Detail Q5)*

Misuse of computer resources is not just physical — it also includes several other forms of improper or unauthorized use.

1. **Unauthorized access:** Using computers, computer software, computer data, or networks without permission.
2. **Improper use:** Using computer resources for a purpose other than the one they were allowed for.
3. **Illegal use:** Breaking (violating) any software licence or copyright.
4. **Interfering with others:** Harassing or threatening other users, or disturbing their access to computing facilities.
5. **Improper alteration of system files:** Changing accounting system files or audit trails without permission, in order to change or delete records.

### Securing information privacy and confidentiality (5.10)

1. **Privacy** means a person's right to control how their personal information is collected, used, and shared.

2. **Confidentiality** means how much information can be kept secret.

Not knowing how to protect our personal information online can cost us a lot: stress, or worse, if the wrong people get hold of our personal data.

**Important tips to protect your privacy:**

1. **Pick strong and unique passwords:** Do not use easy passwords linked to your personal life. Mix capital letters, numbers, and symbols. Use a different password for each account.
2. **Use a VPN:** A **VPN (Virtual Private Network)** adds an extra layer of privacy by encrypting (scrambling and protecting) the data that leaves your computer, and sending it through other (third-party) servers.
3. **Avoid clicking links and attachments in emails:** Phishing emails often tell sad stories, or have strange links, to trick you into giving your details. Scan any attachment with anti-malware software first.
4. **Log out of websites properly:** If you don't log out, you leave the door open for others to use your account, like leaving your wallet in a shop. If a laptop with automatic logins is stolen, the thief can easily see your private data. Use a password manager instead of auto logins.
5. **Use your browser's incognito mode:** In this mode, the next person can't see your browsing history. It also deletes cookies and temporary files that track what you do online.
6. **Keep your anti-malware up to date:** Trojan horses and worms (types of harmful software) can infect a computer and send everything you type to criminals. Updated anti-malware software stops them.
7. **Be careful what you share online:** Instagram and Twitter are hotspots for phishers and cybercriminals. Do not publicly share your identity, your location, or your date of birth.

## TOPIC 7: POSSIBLE DANGERS OF THE INTERNET (5.11)

The internet is a great place to find information, but it is also full of dangers. Many users do not even notice these dangers.

1. **Depression and anxiety:** Spending too much time on social networking sites can badly affect your mood. People who use social media too much are more likely to have poor mental health, including anxiety and depression.
2. **Cyberbullying:** Bullying people using electronic means is called cyberbullying, also known as online bullying, and is common among teenagers. *Example:* someone bullying or harassing others on the internet, especially on social media.
3. **FOMO (Fear of Missing Out):** FOMO became common with the rise of social media, and is one of its most common bad effects. People keep posting online because they're afraid others will think their life is boring or not important.
4. **Unrealistic expectations:** Social media creates false expectations about life and friendship, because most of it lacks authenticity (being real and true): people share only exciting adventures and heavily staged (posed) photos.
5. **Unhealthy sleep patterns:** Spending too much time on social media can cause poor sleep. If sleep becomes irregular and work starts to suffer, cut back on social media time. (Sleep happens in stages: Non-REM stages 1, 2, and 3, then REM sleep.)

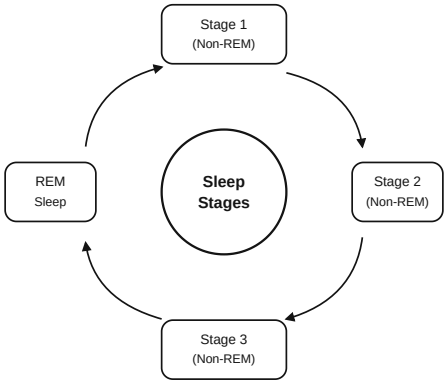



*Fig: Sleep stages cycle (recreated and corrected from the book's Fig. 5.16 — see "Mistakes Found in the Book" below)*

## SHORT QUESTIONS — UNIT 5

1. **Define ethics.** *[Topic 1 · Exercise]*
   Ethics is the study of what is morally good and bad, or right and wrong.
2. **Justify that plagiarism is an offence.** *[Topic 4 · Exercise]*
   Plagiarism means showing someone else's work or ideas as your own. This is cheating, and it is academic dishonesty. So it is a disciplinary offence (a punishable act).
3. **Define phishing in your own words.** *[Topic 2 · Exercise]*
   In phishing, a scammer sends a harmful link. It is made to trick a person into giving away sensitive information, like passwords or bank details.
4. **List the issues of digital ethics.** *[Topic 2 · Exercise]*
   Misuse of personal information, misinformation and deep fakes, lack of oversight and responsibility, and social and political instability.
5. **Discuss any three uses of the internet in business.** *[Topic 3 · Exercise]*
   E-commerce (selling things online, like Amazon), marketing and CRM (serving customers better), and collaboration and document storage (using tools like Google Drive, OneDrive, and Dropbox).
6. **Define CRM.** *[Topic 3 · Exercise]*
   CRM (Client Relationship Management) is software that uses data to help businesses serve customers better, sell more, and find new customers.
7. **Discuss the use of the internet in entertainment.** *[Topic 3 · Exercise]*
   We enjoy YouTube videos, podcasts, music streaming (like Spotify), news sites (like ARY and Geo), and game sites (like Miniclip) on our phone, TV, or computer.
8. **Define copyright in your own words.** *[Topic 4 · Exercise]*
   Copyright is a legal right that protects the original work of its owner, like books, songs, movies, and computer programs.
9. **Differentiate between piracy and plagiarism.** *[Topic 4 · Exercise]*
   Plagiarism means showing someone else's work as your own. Piracy means illegally copying copyrighted material, like books, software, and films.
10. **What is an ethical issue?** *[Topic 1]*
    It is a situation where a moral problem comes up in society or at work.
11. **What is digital ethics?** *[Topic 2]*
    Knowing and following the rules of good behaviour while online is called digital ethics.

12. **What is a fake website scam?** *[Topic 2]*
Cybercriminals make a fake website, usually a shopping site, to get our personal information and misuse it.

13. **What is social networking?** *[Topic 3]*
It is the use of internet-based social media sites to stay connected with friends, family, colleagues, or customers.

## MULTIPLE CHOICE QUESTIONS — UNIT 5

These are the book's own Exercise MCQs plus a few extra practice MCQs, with the correct answer marked ✔ above the option.

1. The ways of online communication involve: *[Topic 2]*

| a) Social media | b) Messengers | c) Chats | ✔ **d) All of these** |
|---|---|---|---|

2. Leaving your devices unprotected can result in: *[Topic 2]*

| ✔ **a) Slow down your computer** | b) Gain money | c) Fast your computer | d) All of these |
|---|---|---|---|

3. In _____, a scammer sends you a malicious link, designed to trick a person into revealing sensitive information. *[Topic 2]*

| a) Fake Websites | b) Tech Support | ✔ **c) Phishing** | d) None |
|---|---|---|---|

4. Fake _____ agents will often ask for login information or remote access to your computer. *[Topic 2]*

| a) Fake Websites | ✔ **b) Tech Support** | c) Phishing | d) None |
|---|---|---|---|

5. Which of the following is NOT a use of the internet? *[Topic 3]*

| a) E-commerce | b) CRM | ✔ **c) SRM** | d) All of these |
|---|---|---|---|

6. CRM stands for: *[Topic 3]*

| a) Client Relationship Model | b) Customer Relation Module | ✔ **c) Client Relationship Management** | d) Client Relation Module |
|---|---|---|---|

7. The term _____ refers to the use of internet-based social media sites. *[Topic 3]*

| ✔ **a) Social networking** | b) Media networking | c) Social websiting | d) Websiting |
|---|---|---|---|

8. _____ is a type of intellectual property right that protects the original work of an owner. *[Topic 4]*

| a) Cyberbullying | b) Plagiarism | c) Piracy | ✔ **d) Copyright** |
|---|---|---|---|

9. Illegally reproducing copyrighted material, such as books, computer programs, and films, is called: *[Topic 4]*

| a) Cyberbullying | ✔ **b) Plagiarism** | c) Piracy | d) Copyright |
|---|---|---|---|

10. Accessing computers, computer software, computer data, or networks without authorization is called: *[Topic 6]*

| ✔ **a) Unauthorized access** | b) Illegal use | c) Authorized access | d) Improper experiments |
|---|---|---|---|

11. Ethics deals with what is morally: *[Topic 1]*

| a) Fast and slow | ✔ **b) Right and wrong** | c) Big and small | d) Old and new |
|---|---|---|---|

12. Aristotle called ethics the science of: *[Topic 1]*

| a) Computers | ✔ **b) Proper behaviour** | c) Numbers | d) Nature |
|---|---|---|---|

13. Knowing and following the rules of being online is called: *[Topic 2]*

✔

a) Media bias **b) Digital ethics** c) Piracy d) Phishing

14. Which of these is a digital ethical issue? *[Topic 2]*

✔

a) 5G **b) Deep fakes** c) Robots d) Smart devices

## Mistakes Found in the Book

1. **Page 110, Answer key, Unit 5 MCQ 9:** The answer key gives (b) plagiarism. But the book itself (page 87) defines “illegally reproducing copyrighted material, such as books, computer programs, and films” as **piracy**, not plagiarism. The correct answer should be (c) piracy. (For your exam, write the official key’s answer, b, if your teacher follows it, though the book’s own text says the correct meaning is piracy.)
2. **Page 93, MCQ 2:** The book letters the four options wrongly as a, c, b, c (two options are both labelled “c”). The correct answer is the first option, “Slow down your computer.”
3. **Page 88:** The book writes “SAO (Service Oriented Architecture),” but the correct short form for this term is **SOA**, not SAO. The same line also calls interfering with others’ computing access “(that’s piracy),” which is incorrect: interfering with or harassing other users is a type of misuse and harassment, not piracy. Piracy means illegally copying copyrighted material.
4. **Page 87:** The book’s definition of piracy says it is “an act of criminal violence.” This wording is incorrect: piracy is a type of theft or copying crime, not an act of violence. The correct description is that piracy is a **criminal act**, with the goal of stealing other people’s ideas and inventions.
5. **Page 91, Figure 5.16 (Sleep Stages):** The sleep stages picture shows “Stage 4” two times (once as REM, once as Non-REM). The four stages should be numbered 1, 2, 3, and 4, with REM sleep coming after them, not relabelled as Stage 4 twice.
6. **Page 86, Uses of internet in entertainment:** The book lists a news channel as “Dunia News.” The well-known Pakistani news channel is spelled “Dunya News.” This looks like a small spelling mistake in the book.
7. **Page 85:** The book uses the full form “Client Relationship Management” for CRM. Many other books call CRM “Customer Relationship Management” instead. For this book’s exam, write **Client Relationship Management**, since that is what this textbook teaches.

# Unit 6 — Entrepreneurship in Digital Age

## TOPIC 1: BUSINESS PLAN

A **business plan** is a written paper that tells people how a business will run. It is one of the most important steps before starting a successful business, no matter who runs it: 1. One person (a **sole trader**). 2. Two or more people together (a **partnership**). 3. A **limited company** (a bigger business owned by shareholders).

### What is a business plan? (6.1, 6.1.1)

1. An important written document.
2. Gives a full, clear picture of where the company is headed.
3. Needed by every business, regardless of size.
4. Lays out the business strategy and main goals.
5. Shows where the business stands now and where it wants to go next.
6. Can help an older company that wants to try a new direction.
7. Works as a **benchmark**, a standard for checking how well the company is doing.

The main areas covered in a business plan are: goals, marketing, competitors, finance, strategy, timeline, operations and team.

### Key sections of a business plan

A business plan has 7 key sections: 1. An executive summary 2. A business description 3. Details of market strategies 4. Competitor analysis 5. A design and development plan of products and services 6. Information about operations and management 7. Financial information, planning and factors

### Long Question: Write a note on the key components of a business plan. [Exercise Brief Q8]

A business plan has seven key components, each covering a different part of the plan, from its summary to its finances.

1. **Executive Summary**: It is a short overview of the whole plan. May include a table of contents, company background, market opportunity, management overview, competitive advantages, and financial highlights.
2. **Business Description and Structure**: It explains why the business exists, what it sells, and how things are made: where materials come from, how stock (inventory) is handled, and how delivery works. For a service business, it explains the service and its value to customers. Also covers legal matters, office details, and expenses.
3. **Competitor Analysis**: It studies competitors (other businesses selling similar things) to plan correctly for demand (what people want) and supply (what we can give).
4. **Development Plan of Products**: It shows how the plan is carried out: the product or service is made through clear steps, with quality checks along the way.
5. **Market Research and Strategies**: It covers the market study and marketing plan: sales forecasts, deadlines, milestones, advertising, and public relations. This is how a business beats its competition. With little data, it can use **testimonials** (comments from real customers) instead.
6. **Management and Personnel**: It gives short bios of the company's managers, showing how their skills help meet business goals. An experienced team reassures investors, who check risk before giving money.
7. **Financial Documents**: It gives the numbers behind the plan: profit and loss statement, balance sheet, and cash flow statement, planned for the next three years.

**Long Question: What is the use of a business plan in the ideology of developing a business? [Exercise Brief Q9]**

A business plan explains a company's goals and strategy, guides its work, and helps it raise money and track its progress.

1. **Clear picture**: It is a written document giving a full, clear picture of the company's future.
2. **Goals and strategy**: It explains the business strategy and main goals, from where the business is now to where it wants to be.
3. **Guide for work**: It acts as a road map: the development plan shows how the product will be made, and the operations plan shows how the business runs day to day.
4. **Getting money**: It lets investors (people who put money into a business) read the plan and financial documents to decide whether to invest.
5. **New direction**: It is useful for an older company looking to change direction.
6. **Benchmark**: It acts as a standard for checking whether the company is doing well.

**Long Question: Why is market research important when planning to set up a business? [Exercise Brief Q10]**

Market research is important because it tells a business about its competitors, demand, and supply, so it can plan correctly and avoid costly mistakes.

1. **Knowing the competitors**: It tells us what competitors sell, at what price, and how.
2. **Demand and supply**: It helps plan correctly for the demand and supply of our product or service.
3. **Marketing strategy**: It helps plan sales forecasts, deadlines, milestones, advertising, and public relations.
4. **Beating competition**: It helps a business compete well and build a successful business plan.
5. **Building trust**: It lets a business use testimonials from real customers instead, when there is little data.

**Conclusion**: Without market research, a business might make something people don't need, or set the wrong price. Market research lowers this risk.

## TOPIC 2: PROMOTION, VALUE PROPOSITION AND QUALITY ASSURANCE

These three ideas help a company sell its products and keep its customers happy.

### Promotion (6.2)

**Promotion** means all the activities that tell people about a product, brand, or service, to make them aware of it, attract them, and make them want to buy it instead of going elsewhere.

1. **Types of promotion**: Advertisements, press releases, and consumer promotions (schemes, discounts, and contests).
2. **Below the line promotions**: Trade discounts, freebies (free gifts), incentive trips, awards, and similar things.
3. Sales promotion is one part of the whole promotion effort.

### Value proposition (6.3)

A **value proposition** is a short, clear statement of the benefits customers get by buying the company's products or services. It works as a promise, both inside the company and in the market.

**Do you know?** The term "value proposition" first appeared in a McKinsey & Co. research paper in 1988. It was defined as: "a clear, simple statement of the benefits, both tangible (things you can touch) and intangible (things you cannot touch), that the company will provide, along with the approximate price it will charge each customer segment for those benefits."

### Quality assurance (6.4)

**Quality assurance (QA)** is a process that helps a business follow the quality standards set by the company or its industry. It is how a company works to improve the quality of its products.

1. **A promise**: Many businesses see QA as a promise to stakeholders (people connected to the business) and customers, to deliver high-quality products and a good experience.
2. **A broad process**: QA stops quality problems before they happen. The QA team checks every stage: production, testing, packaging, and delivery.
3. **Quality control (QC)**: QC is a narrower process than QA. It looks for mistakes, errors, or missing things in the finished product.

**The quality assurance cycle** (from the book's diagram) has 5 steps that repeat in a circle: 1. Identify issues with the process. 2. Generate (create) corrective actions: ways to fix the problem. 3. Verify the corrective action: check that the fix works. 4. Implement the corrective action: apply the fix. 5. Monitor and control: keep watching so the problem does not return.

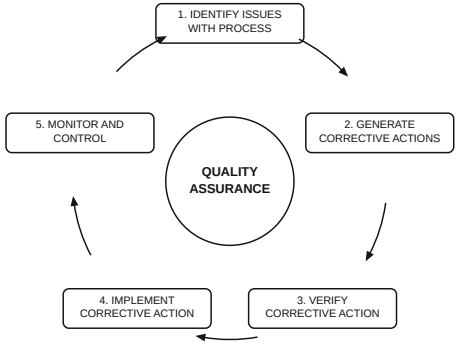



*Fig: The Quality Assurance cycle — 5 steps repeating in a circle*

## TOPIC 3: PROJECT MANAGEMENT AND MEDIA LITERACY

Project management helps a business finish its work on time, while media literacy helps people understand the messages they see in the media.

### Importance of project management and media literacy (6.5)

Handling a project takes a good understanding of scheduling, planning, reporting, and tracking. A good project manager must understand project management well, along with its different job roles.

1. **Project Management**: The Project Management Institute defines it as: "Project management is the application of skills, knowledge, tools and techniques to project activities to meet project requirements." It uses policies, principles and procedures to guide a project from start to finish, giving it leadership and direction.
2. **Media Literacy**: Media literacy helps people understand and take part in public discussions, and make good decisions when choosing leaders, by thinking critically about media messages.

### Long Question: How does project management help in developing a business plan? [Exercise Brief Q7]

Project management helps a business plan succeed by planning, scheduling, tracking, and leading all of its tasks from start to finish.

1. **Meaning**: Project management uses skills, knowledge, tools and techniques to finish project tasks and meet project requirements.
2. **Planning and scheduling**: A business plan has many tasks (market research, product development, finance), and project management plans and schedules all of them.

3. **Reporting and tracking**: It tracks progress and makes reports, so deadlines and milestones in the business plan are met.
4. **Start to finish**: It uses policies, principles and procedures to guide the work from start to completion.
5. **Leadership**: It gives leadership and direction to the team, so everyone knows their job.
6. **Media literacy helps too**: It helps the business understand media messages and public opinion while planning its marketing.

## TOPIC 4: PAYMENTS AND TRANSACTIONS

Payments and transactions happen in every business. They are related ideas, but they are not the same thing.

### Payments (6.6)

A **payment** is when someone freely gives money, or something of equal value, to another person, in exchange for goods or services, or to complete a legal duty.

1. **Payer**: The person who gives the money.
2. **Payee**: The person who gets the money.
3. **Ways to pay**: Cash, cheque, wire transfer, credit card, debit card, or crypto currency (digital money like Bitcoin).

### Transactions (6.7)

A **transaction** is a complete agreement between a buyer and a seller to exchange goods, services, or financial assets (things of money value) for money. The same word is also used in company accounting (keeping financial records).

### Difference between payments and transactions (6.7.1)

**Long Question: Elaborate the difference between a transaction and a payment. [Exercise Brief Q1]**

A payment is the transfer of money or value from a payer to a payee, while a transaction is the complete agreement between a buyer and a seller to exchange goods, services, or assets for money.

| Payment | Transaction |
|---|---|
| It is the transfer of value (money or goods) from one party to another. | It is a complete agreement between a buyer and a seller. |
| It is given for goods or services, or to complete a legal duty. | It is made to exchange goods, services or financial assets for money. |
| It can take many forms: cash, cheque, card, wire transfer, crypto currency. | It is one single instance of buying or selling something. |
| Barter (exchanging one good or service for another) is also a form of payment. | In accounting, events that affect a business's money are recorded as accounting transactions. |
| It involves a payer (who gives) and a payee (who gets). | It involves a buyer and a seller. |

**Example**: When you buy a book from a shop, the whole deal is a transaction. The money you give the shopkeeper is the payment.

## TOPIC 5: DIGITAL MARKETING

**Digital marketing** means using the internet and digital tools to promote a product. It helps companies sell more and build a better brand image (the way people see the company), and has become a key business skill over the last ten years as technology has grown in importance.

## Tools and techniques used in digital marketing (6.8)

### Long Question: Write a note on the tools and techniques used in digital marketing. [Exercise Brief Q2]

Digital marketing uses several online tools and techniques to promote a product and build its brand. There are 9 main techniques:

1. **Social Media Marketing**: It is the most important digital marketing tool today, and a must-have technique for growing a brand digitally.
2. **Search Engine Optimization (SEO)**: Brings more visitors (traffic) to a website by making it appear higher in search results on engines like Google, which also improves brand awareness.
3. **Email Marketing**: Brings customers back to buy new products. People sign up for an email list, often after their first purchase, which builds customer loyalty (keeping customers coming back) and a good relationship with them.
4. **Content Marketing**: It means creating useful, valuable content regularly across different platforms, which builds brand awareness.
5. **Video Marketing**: Works alongside content marketing. Good-quality videos leave a long-lasting image of the brand in people's minds.
6. **Web Advertising**: Brands create clickable advertisements and place them on popular websites.
7. **Affiliate Advertising**: A brand pays a popular person with many followers, usually a blogger or social media influencer, to post about the company.
8. **Overall Personalization**: It means giving a brand its own unique style, which makes people believe the company is successful.
9. **Creating an App**: An app gives customers an easy way to contact the brand, shows them where to buy things, and sends notifications about new products.

**How affiliate marketing works (from the book's diagram):** 1. The advertiser partners with a publisher (like a blogger or influencer). 2. The publisher promotes the product to the customer. 3. The customer buys from the publisher's recommendation. 4. The advertiser pays the publisher.

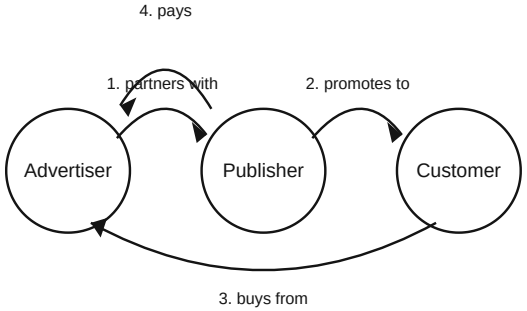



*Fig: How affiliate marketing works*

### Long Question: What role does digital marketing play in today's world? [Exercise Brief Q3]

Digital marketing plays a big role today by increasing sales, building brand image, and keeping customers engaged, mostly through the internet.

1. **Technology in business**: Over the last ten years, technology has become very important in the business world.
2. **More sales**: Digital marketing helps companies improve sales through social media, websites, emails and apps.
3. **Brand image**: It builds a strong brand image, for example through content marketing and video marketing.
4. **Reaching more people**: Using SEO, web advertising and affiliate advertising, a business can reach many people at a low cost.
5. **Keeping customers**: Email marketing and apps bring customers back and keep them updated on new products, which builds loyalty.
6. **Two-way talk**: Customers can talk to the brand easily, and the brand can treat each customer in a personal way.

## TOPIC 6: SEARCH ENGINE OPTIMIZATION AND SOCIAL MEDIA

**SEO** stands for Search Engine Optimization: the process of improving a website so it is easier to find in search results.

### Search engine optimization (6.9)

SEO makes it easier for people to find a website when they search for products or services on Google, Bing, and other search engines. The easier a website is to find, the more likely it attracts new and existing customers.

### Long Question: How does search engine optimization help the business grow? [Exercise Brief Q4]

Search engine optimization helps a business grow by making its website easier to find, which brings in more customers and more profit.

1. **Better visibility**: SEO improves a website so it appears higher in search results on Google, Bing, and other search engines.
2. **More customers**: Better visibility means more attention for the business, attracting both new (prospective) and existing customers.
3. **Huge number of searches**: People search online trillions of times every year, often to find products and services they want to buy.
4. **Main source of traffic**: Search is often the main way people find a brand online, and it also supports other marketing methods.
5. **Beating competitors**: Ranking higher than competitors in search results has a big effect on the business's profit.
6. **Brand awareness**: SEO helps more people become aware of the brand.

**The SEO Pyramid** (from the book's diagram) has 4 levels, building up from a strong base: 1. **Accessible, quality content** (the base, built first): It means having unique text content, content that search engine "bots" can read, a clear URL structure, internal links, sitemaps, and working server response codes. 2. **Keyword research and targeting**: It means finding and choosing the right keywords, and placing them well in titles, meta tags, URLs, and headings (H1s). 3. **Link building**: It means asking for links and creating content that naturally earns links from other websites. 4. **Social** (the top level): It means getting users to engage on the site, using social media and viral marketing (content that spreads quickly online).

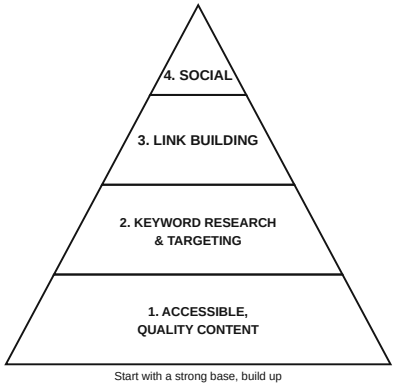



*Fig: The SEO Pyramid — 4 levels, base to top*

### Twitter (6.9.1)

1. Twitter came from a podcasting company called Odeo, started in 2004 by Evan Williams, Biz Stone and Noah Glass.
2. In 2005, Apple added podcasts to iTunes, so Odeo needed a new direction.
3. Engineer Jack Dorsey suggested a short message service (SMS) so friends could share small, blog-like updates, and Glass suggested the name "Twttr".

4. Dorsey sent the first tweet (“just setting up my twttr”) on March 21, 2006. The full version of Twitter started in July 2006.
5. In October 2006, Williams, Stone and Dorsey bought Odeo and started a new company called Obvious Corp.

**Snapchat (6.9.2)**

1. Evan Spiegel and Bobby Murphy met at Stanford University, and in spring 2011, they made Picaboo, the first version of Snapchat.
2. It was an app that sent photos which disappeared after a short time, and it launched on the iOS App Store in July.
3. Users found they could take screenshots, making the disappearing photos pointless, so the creators added a notification for when someone takes a screenshot.
4. In September, they renamed the app Snapchat, added captions to photos, and relaunched it, focusing more on improving the technology than on advertising.

**Blogs (6.9.3)**

A **blog** (short for “weblog”) is an online journal or information website that shows posts in reverse order by time, so the newest one appears first, at the top. A writer, or a group of writers, uses it to share their views with people all over the world.

## TOPIC 7: TECHNOLOGY AND DIGITAL PLATFORMS FOR ENTREPRENEURSHIP

An **entrepreneur** is a person who starts and runs a new business. **Entrepreneurship** is the work of starting and growing one, often by trying a new idea. Technology plays an important role in helping entrepreneurship grow today.

**Technology as an enabler in entrepreneurship (6.10)**

**Entrepreneur enabling** means working with and supporting people who want to start businesses. **Entrepreneurship enablers** are the things that make it possible for them to start and grow; technology is one of the biggest.

1. Technology improves **infrastructure** (basic systems like the internet and payment systems) and business culture.
2. **ICT** (Information and Communication Technology) acts as an **enabler**: it gives entrepreneurs new opportunities, but can also create some limits (constraints).
3. ICT also acts as an **implementer**: it gives modelling tools and systems engineering to help build new business processes.
4. Together, ICT as an enabler and as an implementer create a **new business process**.
5. Old businesses are becoming digital, and new digital businesses are being created. Together this is called **digital entrepreneurship**, which brings innovation and helps the economy grow.
6. Digital platforms like Facebook, Instagram, Amazon, and PayPal let people sell, market, and receive payments easily.

Digital technology connects three things: what makes a business possible (the **enabler**), what it achieves (the **outcome**), and the situation around it (the **context**).

**Long Question: How does technology promote entrepreneurship in the economy? [Exercise Brief Q5]**

Technology promotes entrepreneurship by acting as both an enabler and an implementer, giving entrepreneurs the infrastructure, tools, and digital platforms to start and grow a business.

1. **Enablers**: Entrepreneurship enablers make it possible for entrepreneurs to start and grow, and technology is one of the biggest.
2. **Infrastructure and culture**: Technology improves infrastructure (like the internet and payment systems) and business culture.
3. **ICT as an enabler**: ICT gives opportunities to entrepreneurs, but it can also create some limits.
4. **ICT as an implementer**: ICT gives modelling tools and systems engineering to build new business processes.

5. **Digital entrepreneurship**: Old businesses are becoming digital, and new digital businesses are being created, bringing innovation and economic growth.
6. **Digital platforms**: Platforms like Facebook, Instagram, Amazon, and PayPal let people sell, market, and receive payments easily.

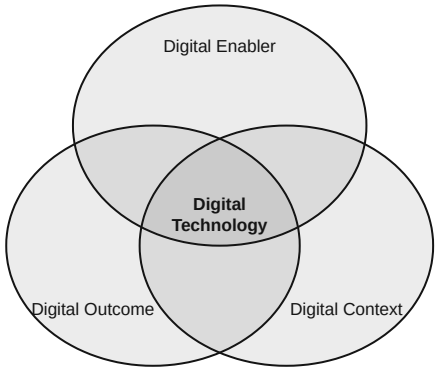



*Fig: Digital Enabler, Digital Outcome and Digital Context overlap into Digital Technology*

**Digital platforms used for entrepreneurship (6.11)**

Entrepreneurship is a main driving force behind innovation and economic growth. Digital entrepreneurship is rising fast in two ways: old businesses going digital, and new digital businesses being created.

A **Digital Business Platform** combines different technologies to create new business abilities in creative ways. It forms a logical layer that separates the business plan from the technology running behind it.

**Long Question: State and explain the types of digital platforms used for entrepreneurship. [Exercise Brief Q6]**

Digital platforms exist at many levels: some are high-level platforms supporting a whole business model, while others are low-level, giving business or technology abilities to other products. The most common ones are:

1. **Social media and media-sharing platforms**: These let people share posts, photos, and videos. Examples: Instagram and YouTube.
2. **Knowledge platforms**: People ask questions and share knowledge here. Examples: Quora and Reddit.
3. **Service-oriented platforms**: These connect people who offer a service with people who need it. Example: Airbnb.
4. **E-commerce platforms**: These are used to buy and sell goods online. Example: Alibaba.

**Types of platform business models (from the book's chart):**

| Platform type | Examples |
|---|---|
| Social Media | Facebook, LinkedIn, Twitter |
| Communication | WhatsApp, Skype, WeChat, LINE |
| Search | Google, Bing, Yahoo, Safari |
| Sharing Economy | Airbnb, Uber (peer-to-peer), ZipCar; file and music sharing |
| Matching | TaskRabbit, Tinder, e-Harmony |
| Development | App stores, gaming consoles |

| Platform type | Examples |
|---|---|
| Content and Review | YouTube, TripAdvisor, Yelp |
| Booking Aggregators | Booking.com, Expedia, Agoda |
| Retail | Amazon, Etsy, eBay, Craigslist |
| Payment | PayPal, Alipay, Visa |
| Crowdsourcing and Crowdfunding | InnoCentive, Kickstarter |

A **sharing economy platform** lets people share things instead of buying new ones. The book explains four kinds: 1. Sharing of limited assets and resources, like cars and bicycles. 2. Peer-to-peer sharing (person to person), like Airbnb and Uber. 3. Sharing of assets the platform owner provides, like ZipCar. 4. Sharing of resources that are not limited, like files, music, and information. This is mostly peer-to-peer.

**Crowdfunding** means raising small amounts of money from many people, usually online, to fund a new idea or project. InnoCentive and Kickstarter are examples of crowdsourcing and crowdfunding platforms.

## SHORT QUESTIONS — UNIT 6

1. **Elaborate the difference between a transaction and a payment.** *[Topic 4 · Exercise]*
   A payment is the transfer of money or value from a payer to a payee. A transaction is the complete agreement between a buyer and a seller to exchange goods or services for money.
2. **Write the names of tools and techniques used in digital marketing.** *[Topic 5 · Exercise]*
   Social media marketing, SEO, email marketing, content marketing, video marketing, web advertising, affiliate advertising, overall personalization, and creating an app.
3. **What role does digital marketing play in today's world?** *[Topic 5 · Exercise]*
   It improves sales and brand image. It reaches many people at a low cost. It also brings customers back through emails and apps.
4. **How does search engine optimization help the business grow?** *[Topic 6 · Exercise]*
   SEO makes a website appear higher in search results. This means more customers see it and visit it. This increases sales and profit.
5. **How does technology promote entrepreneurship?** *[Topic 7 · Exercise]*
   Technology makes it possible for entrepreneurs to start and grow. It gives opportunities, tools, and digital platforms for selling and getting paid.
6. **State the types of digital platforms.** *[Topic 7 · Exercise]*
   Social media platforms (Instagram, YouTube), knowledge platforms (Quora, Reddit), service-oriented platforms (Airbnb), and e-commerce platforms (Alibaba).
7. **How does project management help in developing a business plan?** *[Topic 3 · Exercise]*
   It plans, schedules, reports and tracks the tasks of the plan. It guides the work from start to finish.
8. **What is the use of a business plan?** *[Topic 1 · Exercise]*
   It explains the business strategy and goals, guides the work, helps get money from investors, and acts as a benchmark for performance.
9. **Why is market research important when setting up a business?** *[Topic 1 · Exercise]*
   It tells us about competitors, demand, and supply. This helps us make a good marketing strategy and beat the competition.

10. **Define business plan.** *[Topic 1 · Book concept]*
A business plan is an important written document. It gives a detailed description and full overview of a company's future.

11. **Name the seven key sections of a business plan.** *[Topic 1 · Book concept]*
Executive summary, business description, market strategies, competitor analysis, product development plan, operations and management plan, and financial information.

12. **What is included in an executive summary?** *[Topic 1 · Book concept]*
Table of contents, company background, market opportunity, management overview, competitive advantages, and financial highlights.

13. **What is promotion?** *[Topic 2 · Book concept]*
Promotion is the whole set of activities that tell people about a product, brand, or service, to make them want to buy it.

## MULTIPLE CHOICE QUESTIONS — UNIT 6

These are the book's own Exercise MCQs plus a few extra practice MCQs, with the correct answer marked ✔ above the option.

1. A __ is an essential written document that provides a detailed description of a company's future. *[Topic 1 · Exercise MCQ 1]*

| a) Quality assurance | b) Entrepreneurship | ✔ **c) Business plan** | d) Project management |
|---|---|---|---|

2. __ may include a table of contents, company's background, market opportunity and financial highlights. *[Topic 1 · Exercise MCQ 2]*

| a) Business description | b) Competitors analysis | c) Financial documents | ✔ **d) Executive summary** |
|---|---|---|---|

3. __ can be defined as a set of activities which communicates the product, brand or services to the user. *[Topic 2 · Exercise MCQ 3]*

| a) Value proposition | ✔ **b) Promotion** | c) Quality assurance | d) Entrepreneurship |
|---|---|---|---|

4. A __ is a completed agreement between a buyer and seller to exchange goods, services or financial assets for money. *[Topic 4 · Exercise MCQ 4]*

| a) Payment | ✔ **b) Transaction** | c) Financial statements | d) Contract |
|---|---|---|---|

5. __ is the digital marketing technique that involves creating more traffic to a website by making sure the website appears higher up in search results. *[Topic 5 · Exercise MCQ 5]*

| ✔ **a) Search engine optimization** | b) Email marketing | c) Social media marketing | d) Content marketing |
|---|---|---|---|

6. A __ is an online journal or information website displaying information. *[Topic 6 · Exercise MCQ 6]*

| a) Tweet | b) Snapchat | ✔ **c) Blogs** | d) Facebook profile |
|---|---|---|---|

7. A __ in marketing is a concise statement of the benefits that a company is delivering to the customers who buy its product or services. *[Topic 2 · Exercise MCQ 7]*

| a) Competitors analysis | b) Quality assurance | c) Promotion | ✔ **d) Value proposition** |
|---|---|---|---|

8. __ includes the sales forecast, deadlines and milestones, advertising, public relations to formulate a successful business plan. *[Topic 1 · Exercise MCQ 8]*

| a) Financial document | b) Management and personnel | ✔ **c) Market research and strategies** | d) Development plan of your product |
|---|---|---|---|

9. __ enables the populace to understand and contribute to the public discourse, and eventually make sound decision when electing their leaders. *[Topic 3 · Exercise MCQ 9]*

| a) Project management | ✔ **b) Media literacy** | c) Quality assurance | d) Social media marketing |
|---|---|---|---|

10. __ is the voluntary transfer of money, equivalent, or other valuable items from one person to the other person in exchange of goods and items. *[Topic 4 · Exercise MCQ 10]*

| ✔ **a) Payments** | b) Transactions | c) Barter | d) Contracts |
|---|---|---|---|

11. How many key sections does a business plan have? *[Topic 1]*

| a) five | b) six | ✔ **c) seven** | d) nine |
|---|---|---|---|

12. Profit and loss statement, balance sheet and cash flow statement are part of: *[Topic 1]*

| a) executive summary | ✔ **b) financial documents** | c) competitor analysis | d) promotion |
|---|---|---|---|

13. Financial projections in a business plan are made for the next: *[Topic 1]*

| a) one year | ✔ **b) three years** | c) ten years | d) six months |
|---|---|---|---|

14. Trade discounts, freebies and awards are __ promotions. *[Topic 2]*

| ✔ **a) below the line** | b) above the line | c) online | d) value |
|---|---|---|---|

## Mistakes Found in the Book

1. **Answer key swaps B and C for four MCQs.** The book's printed Unit 6 answer key (checked by the task) gives: 1-B, 2-D, 3-C, 4-B, 5-A, 6-C, 7-D, 8-B, 9-C, 10-A. But when we check each MCQ's own options printed on pages 107–108, the correct answers for MCQs 1, 3, 8 and 9 are actually option **C, B, C and B** — not B, C, B and C as the key says. For example, MCQ 1 asks for the word meaning "an essential written document… of a company's future," and option (c) on that page is "Business plan" — the only correct choice — yet the key marks it (b). These notes mark the ✔ on the option that matches the book's own teaching text, which is the one a student should pick.
2. **Snapchat's first name is given two different ways in the same paragraph.** On page 103, the book first says Spiegel and Murphy "created Peekaboo, the first version of Snapchat," and then two sentences later calls it "the initial Picaboo app." The app's real first name was **Picaboo**, so "Peekaboo" is a typing mistake by the book.
3. **"Evan William" should be "Evan Williams."** On page 102, the book spells the Twitter co-founder's name as "Evan William" (missing the final "s"). His correct name is Evan Williams.
4. **"Pagoda" should be "Agoda," and "Safari" is wrongly listed as a search platform.** In the "Types of Platform Business Models" chart on page 105, the Booking Aggregators example list prints "Pagoda" — this should be "Agoda" (a real booking website). The Search Platforms box lists "Safari" alongside Google, Bing and Yahoo — but Safari is a web browser, not a search engine, so this is a mistake in the book's own chart.
5. **No "Answer in detail" section in this unit's Exercise.** Unlike some other units, Unit 6's Exercise (pages 107–108) has only 10 "Briefly Answer" questions and no separate long-answer section. These notes still answer all 10 of them in full, numbered detail (as Long Questions) and also give them in short form, so nothing from the Exercise is left incomplete.